

Introduction to Pharmacy (Phar2071)

(2 EtCTS)

Dawit Teshome
Department of Pharmaceutics and Social
pharmacy, SoP, AAU

Introduction

► What is Health?

► *‘Health is a state of complete **physical, mental and social** well-being and not merely the **absence of disease** or infirmity.’*
(WHO, 1946)

► **Physical health:** the maintenance of physiological homeostasis through changing circumstances

► e.g., blood glucose level, blood salinity, blood pressure, core body temperature

► **Mental health:** capacity to cope, recover from strong psychological stress, and prevent post-traumatic stress disorders

► **Social health:** capacity to fulfil their potential and obligations, the ability to manage their life with some degree of independence despite a medical condition, and the ability to participate in social activities including work

Introduction...cont'd

- ▶ The human right to Health
 - ▶ Fundamental **moral principles and the legal obligations** of states to ***respect, protect*** and ***fulfil*** human rights, including the right to health.
 - ▶ These obligations derive **from international law, regional human rights agreements and national laws**
 - ▶ **Access** to health care and essential medicines are also human right
 - ▶ ***Pharmacy practice*** is one of the component in health care environment which aims to improve health.
- ▶ **Medicinal therapy** is the most frequently used form of treatment

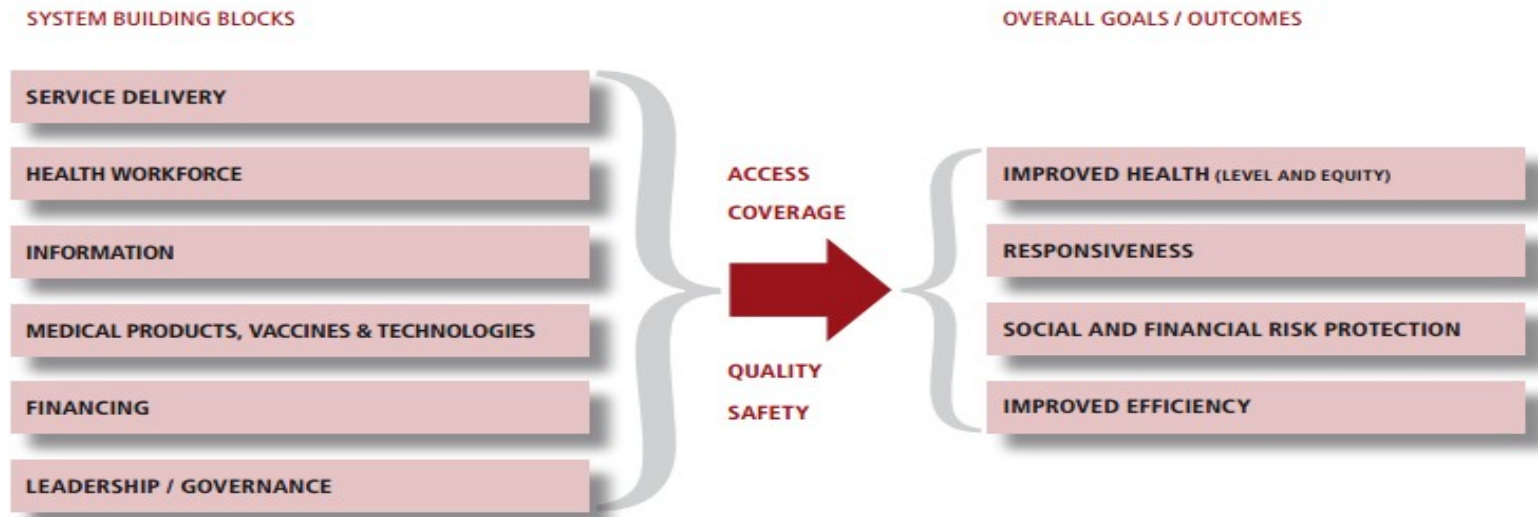
Intro.....cont'd

- **Medicinal therapy** use has grown dramatically. The reasons are:
 - Having more **aged** population
 - Growing prevalence of **chronic diseases**
 - **Emerging new infectious** diseases
 - Increasing **range of effective** medications
 - More and more so-called “**life-style medicines**”
 - Dry skin, wrinkles, Erectile Dysfunction etc

1.1 Introduction to the healthcare system in Ethiopia

- ▶ A health system consists of all the organizations, institutions, resources and people whose primary purpose is to improve health.
- ▶ According to WHO, the six core “building blocks” of a health system are:

THE WHO HEALTH SYSTEM FRAMEWORK



- ▶ Who is financing, providing and regulating healthcare services?
 - ▶ State
 - ▶ non governmental (societal-based and/or private actors)
 - ▶ market

1.1 Ethiopian Health care system

- ▶ **Pluralistic**: traditional and modern
 - ▶ WHO estimate: TM 70 - 80%
- ▶ **Medical pluralism**: refers to the employment of more than one medical system or the use of **both conventional and complementary and alternative medicine** (CAM) for health and illness

1.1.1 Modern Medicine in Ethiopia

- ▶ Introduced in 16th century in fragmented way
 - ▶ Emperor Libenedengel
- ▶ Further progress in the introduction and development- reign of **Emperor Menilek** (1889 – 1913)
 - ▶ Russian Red Cross Hospital – 1897 in Addis Ababa

Modern Medicine in Ethiopia

- ▶ Establishment of **health department** within **Ministry of Interior**- 1908
- ▶ The **first** governmental health facility established in **Harar** by Ras Mekonnen and Menelik II Hospital in 1909
- ▶ 1935- health manpower **training** started by enrolling students for **auxiliary** medical training
- ▶ Italian invasion – **interrupt** the activity
- ▶ 1947 – 1st health legislation

1.1.2 Structure of Health care services

- *Modern medical care model (MMC)*
 - Earlier period of introduction of health services
 - **Urban** centered and **Hospital** based
 - Services were **inaccessible** to majority of rural population due to
 - geographical, economical and cultural barrier
 - It was ***mechanistic*** approach and not holistic

Approach of Health care services

▶ *Basic Health Service Approach (BHSA)*

- ▶ Dominant theme in **50's** and **60's**
- ▶ Criticism against **mechanistic** approach
- ▶ Success stories utilizing large number of **medical auxiliaries** in China
- ▶ Reframed and shifted utilization of **medical** auxiliary from medical facility to **health center** and **health institution**
- ▶ Principle ” *taking services where people live*” through home visits, outreaches, prison health services by medical auxiliaries

Approach of Health.....cont'd

- ▶ The era of BHSA in Ethiopia
 - ▶ One of the **success** stories registered in health sector development
- ▶ The organization and development of health services at that time can be considered as **four** tiered system:
 - ▶ Health station (with health assistant at grass root level)
 - ▶ Health center at Awraja level (health center team)
 - ▶ Regional referral Hospital (Tekelaygizat level)
 - ▶ Central referral hospital

Approach of Health.....cont'd

- ▶ Problem with BHSA
 - ▶ Constrained with respect to **facility** and **manpower**
 - ▶ **Socio-cultural** factor prevent population from using available sources
 - ▶ No mechanism for **community participation** that can ensure sustainability and culturally acceptable care
- ▶ *Primary Health Care (PHC) approach*
 - ▶ WHO and UNICEF re-examined their practice and this led to **Alma-Ata** declaration in 1978

Approach of Health.....cont'd

- Primary health care (PHC)
 - **the first level** of contact individuals, families and communities have with the **health care system**.
- Primary Health Care defined as
 - Essential health care based on **practical**, scientifically **sound** and **socially** acceptable methods and technology made **universally accessible** to individuals and families in the **community** through their **full** participation and at a **cost** that the community and the country can **afford** to maintain at every stage of their development in the spirit of self-reliance and self-determination”

The five principles usually associated with the PHC strategy

- **Accessibility**
 - Primary care services are available, affordable
 - Provided equally to all individuals irrespective of their gender, age, ethnicity or location.
- **Public or community participation**
 - involving all of community's resources in promoting health and addressing health problems at the grass roots level.
- **Health promotion**
 - Helping a community to strengthen the socioeconomic conditions that contribute to good health.

Approach of Health.....cont'd

- ▶ Component/element of PHC
 - ▶ Immunization
 - ▶ Food supply and proper nutrition
 - ▶ Water and sanitation
 - ▶ Appropriate treatment of common disease and injuries
 - ▶ Maternal child birth care including family planning
 - ▶ Providing essential drugs
 - ▶ Mental health

Approach of Healt....cont'd

- ▶ Education concerning the prevailing health problems and methods of preventing and controlling them
- ▶ Provision and control of locally endemic
- ▶ Dental health
- ▶ Traditional medicine
- ▶ Control of HIV/AIDS and STIs
- ▶ Occupational health
- ▶ As a member of WHO, Ethiopia signed the *Almata* charter
- ▶ WHO internationally evaluated PHC strategy (1984 – 1985)

Approach of Health.....cont'd

▶ Problems observed:

- ▶ Very limited intersectoral collaboration
- ▶ Lack of community involvement due to lack of remuneration of trained CHWs; 85% were found to be non functional

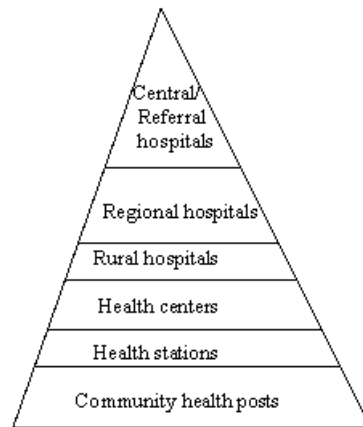
▶ Achievement:

- ▶ Expansion of health services
- ▶ Control of iodine deficiency disorder
- ▶ Training in health education

Approach of Health.....cont'd

- Organization of health delivery system was six (five) tiered
 - **Community health service (not formal) e.g. traditional birth attendants**
 - Health station
 - Health center
 - Zonal Hospital
 - Regional referral hospital
 - Central referral and training hospitals

Six-tier health care delivery system in Ethiopia



Approach of Health.....cont'd

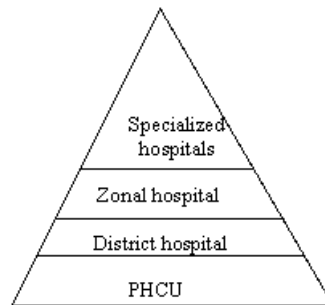
- Former Health care structure before BPR- **4 tier**
 - *Primary health care unit PHCU (HC+5 HP)*
 - 10 beds, Technical staffs (13 HC+15 HP)
 - Population served – 25,000
 - *District/zonal hospital*
 - 50 beds, Technical staffs -33
 - Population served – 1,000,000

Approach of Health.....cont'd

– Specialized hospitals

- 250 beds
- Technical staffs – 120
- Population served – 5,000,000

Four -tier health care delivery system



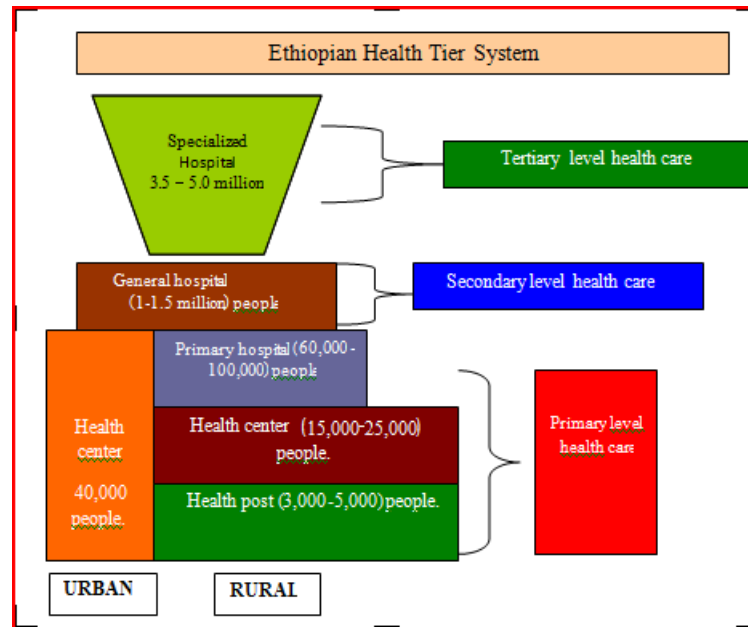
Approach of Health.....cont'd

- Current health care delivery system : a *three-tier*
- *First level of a Woreda/District health system*
 - Comprising a primary hospital
 - population coverage of 60,000-100,000 people),
 - Health centers (1/15,000-25,000 population)
 - Satellite Health Posts (1/3,000-5,000 population)
- Connected to each other by a referral system

Approach of Health.....cont'd

- *NB! Primary Hospital, Health center and health posts form a Primary health care unit (PHCU) with each health center having five satellite health posts.*
- *Second level - General Hospital*
 - Population coverage of 1-1.5 million people
- *Third level - a Specialized Hospital*
 - Population of 3.5-5 million

Three-tier health care delivery system in Ethiopia (currently used)



1.2. Scope of Pharmacy: Education and Careers

- ▶ What is pharmacy?
 - ▶ Pharmacy is the *art* and *science* of preparing and dispensing medications and the provision of drug related information to the public.
- ▶ It involves
 - ▶ Interpretation of prescription orders;
 - ▶ Compounding, labeling and dispensing of drugs and devices;
 - ▶ Drug product selection and drug utilization reviews;
 - ▶ Patient monitoring and intervention
 - ▶ The provision of cognitive services related to use of medications and devices

Scope of Pharmacy.....cont'd

- Currently medicines can be
 - purchased in new settings (supermarkets, drug stores or at markets)
 - handled by non-pharmacists,
 - obtained by mail order or over the Internet,
 - sold by medical practitioners and
 - dispensed by computerized dispensing machines.
- Compounding has been largely replaced by the commercial manufacture of nearly all formulations.
- **If so, why do we need a pharmacist?**



Scope of Pharmacy.....cont'd

- A **paradigm shift** in pharmacy practice is occurring
 - From **product** oriented to **patient** oriented
 - Change in role of pharmacists - “**drug therapy manager**”

- **Pharmaceutical care:**

*“the responsible provision of **drug therapy** for the purpose of achieving definite **outcomes** that improve a patient’s quality of life”.*

Pharmacy Education

- ▶ Entry level degree programs
 - ▶ B Pharm
 - ▶ Pharm D
- ▶ Courses include
 - ▶ Social sciences (e.g. Psychology, logic, economics, accounting, management)
 - ▶ Physical and biological sciences (e.g. organic chemistry, anatomy, biochemistry, physiology)
 - ▶ Professional courses
- ▶ Therapeutic outcomes:
 - ▶ Cure of a disease,
 - ▶ Elimination or reduction of a patient's symptoms,
 - ▶ Arresting or slowing a disease process or symptoms

- ▶ The seven roles and responsibilities of Pharmacists are:-
 - ▶ Care giver
 - ▶ Decision maker
 - ▶ Communicator
 - ▶ Leader
 - ▶ Manager
 - ▶ Life-long learner
 - ▶ Teacher
- ▶ Pharmacists must possess specific knowledge, attitudes, skills and behaviours to equip them to perform these roles effectively

► The following are relevant areas of study :-

- **biological systems**, the chemistry of drugs and other constituents of medicines, pathophysiology and disease states and the interaction between medicines and biological systems, · dosage form design and development
- the **actions** and **uses** of medicines and other relevant products,
- the **laws** governing the practice of pharmacy and the sale and supply of medicines
- the principles governing **ethical** conduct and the relevant national **Code**
- safety and risk management
- pharmaco-epidemiology and pharmaco / health-economics
- **practice** of pharmacy in community and hospital pharmacies, industrial, academic, and where appropriate, clinical biology
- the relevant aspects of **the social and behavioural** sciences, leading to competency in delivering patient care
- effective management of **resources** (human, physical, fiscal and time)

▶ **Professional courses**

- ▶ Pharmacognosy
- ▶ Medicinal chemistry
- ▶ Pharmacology
- ▶ Pharmaceutics
- ▶ Biopharmaceutics
- ▶ Clinical pharmacy internships.
- ▶ Pharmaceutical analysis
- ▶ Pharmacotherapeutics
- ▶ Social and administrative pharmacy
- ▶ Pharmacy ethics and Jurisprudence

► Pharmacognosy

- the study of drugs of natural origin.
- the study of the **physical, chemical, biochemical and biological properties of drugs**, drug substances or potential drugs or drug substances of **natural origin** as well as the search for new drugs from natural sources".

► Pharmacology

- a science that studies **drug action** (how medicines and other drugs work and are processed by the body).

- **Medicinal chemistry**

- A disciplines at the intersection of chemistry, especially synthetic organic chemistry, and pharmacology and various other biological specialties, where they are involved with **design, chemical** synthesis and development for **market** of pharmaceutical agents, or bio-active molecules (drugs).

► **Pharmaceutics**

- A discipline of pharmacy that deals with the process of **turning a new** chemical entity (NCE) or old drugs into **a medication** to be used **safely** and **effectively** by patients. It is also called the science of dosage form design.

- **Biopharmaceutics**

- The study of the physical and chemical properties of a drug, and its dosage form, as related to the **onset, duration, and intensity** of drug action

- ▶ Clinical pharmacy
 - ▶ a branch of pharmacy practice that **emphasizes the therapeutic use** of drugs rather than the preparation and dispensing of drugs.
- ▶ Pharmaceutical analysis
 - ▶ is a branch of practical chemistry that involves a series of process for **identification, determination, quantification** and **purification** of a substance, separation of the components of a solution or mixture, or determination of structure of chemical compounds.

- Pharmacotherapeutics
 - 1. Study of the uses of drugs in the treatment of disease.
 - 2. The study of the use of drugs as therapeutic agents, their administration, pharmacokinetics, and effectiveness, and their side effects and toxicity.

Pharmacy Education cont...

- *Graduate education:*
 - Pharmaceutics,
 - Pharmacology,
 - Pharmaceutical/medicinal chemistry,
 - Pharmacognosy, and
 - Social and administrative pharmacy.

Careers in Pharmacy

- In many of the cases the job opportunities, working conditions, job satisfaction and monetary benefits are **excellent**.
- **Job opportunities** are expected to grow b/c of:
 - The increase in **chronic diseases**
 - The likelihood of scientific advances - **more drug products**
 - **New developments** in administering medication
 - Increasingly **well-informed consumers**

▶ *Community Pharmacy*

▶ *Health Systems Pharmacy*

- ▶ private and government-owned **hospitals**,
- ▶ clinics, and other facilities

▶ *Nuclear Pharmacy*

- ▶ Applies the principles and practice of pharmacy and nuclear chemistry to produce **radioactive** drugs used for **diagnosis** and **therapy**.

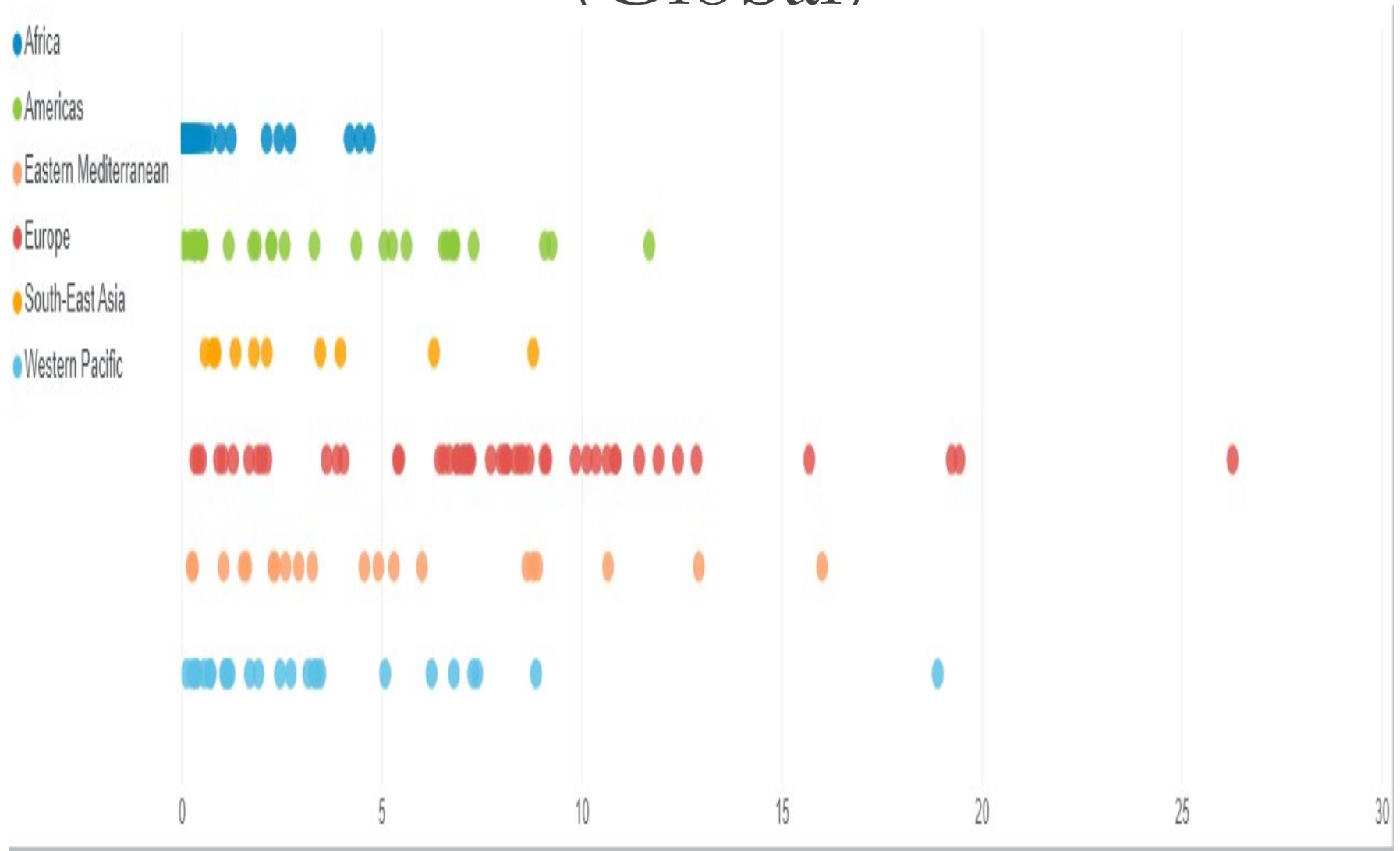
▶ *Industrial pharmacy*

- ▶ **cosmetic, medical device, and biological** products industries

Careers in Pharmacy cont...

- ▶ *Government service* - armed forces, FDA
- ▶ *Academia:*
- ▶ *Pharmaceutical Journalism*
- ▶ *Organizational management* - national and state associations and on boards of pharmacy
- ▶ **Regulatory Affairs:**
- ▶ *Consultancy:* for highly technical and experienced pharmacists

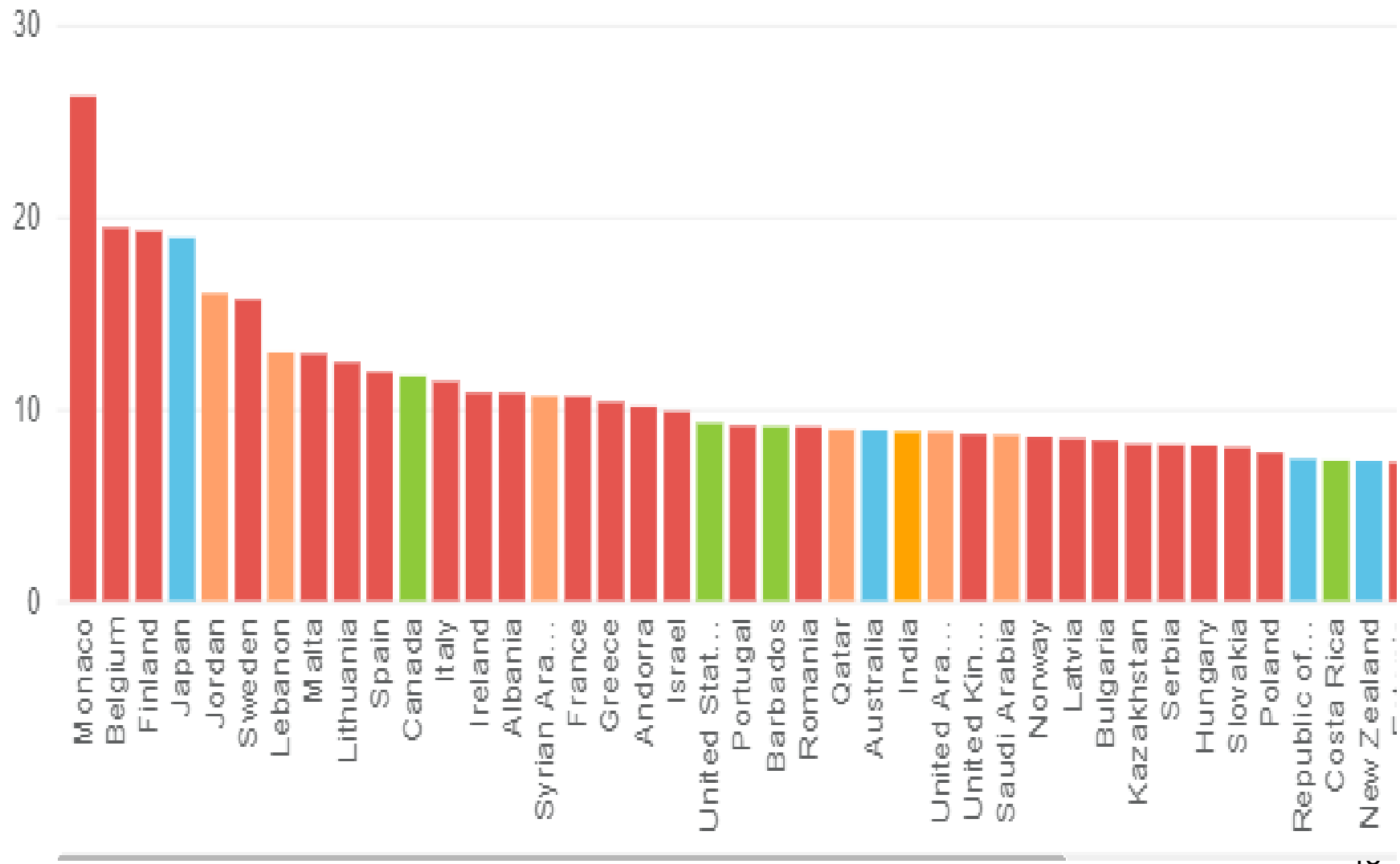
Pharmacists (per 10 000 population) (Global)



Pharmacists (per 10 000 population) (Global)



Pharmacist distribution by country



How many pharmacists are there

1 1 11 2

In **74** countries and territories
(representing **5,614** million people,
76% of the world's population),
there are

4,067,718

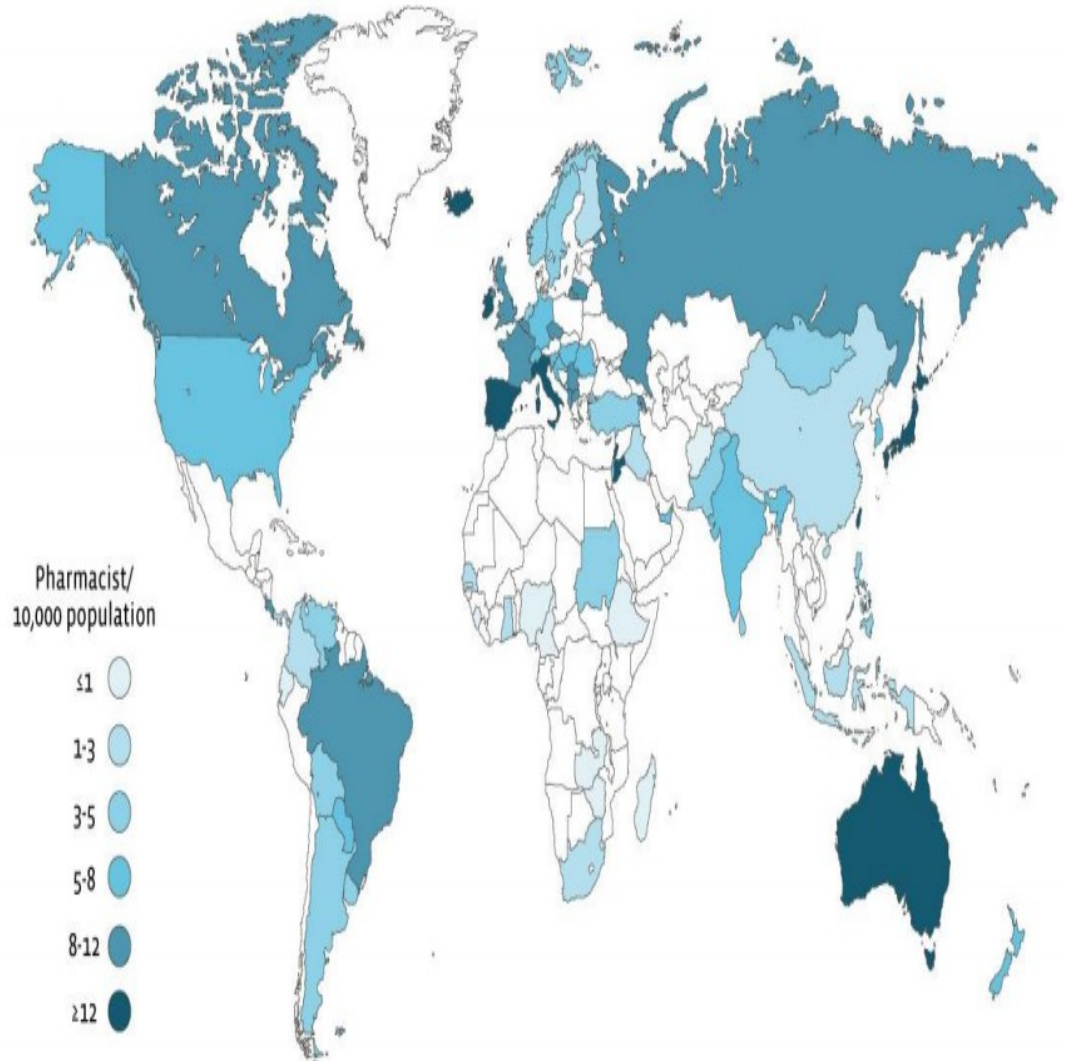
licenced or registered pharmacists



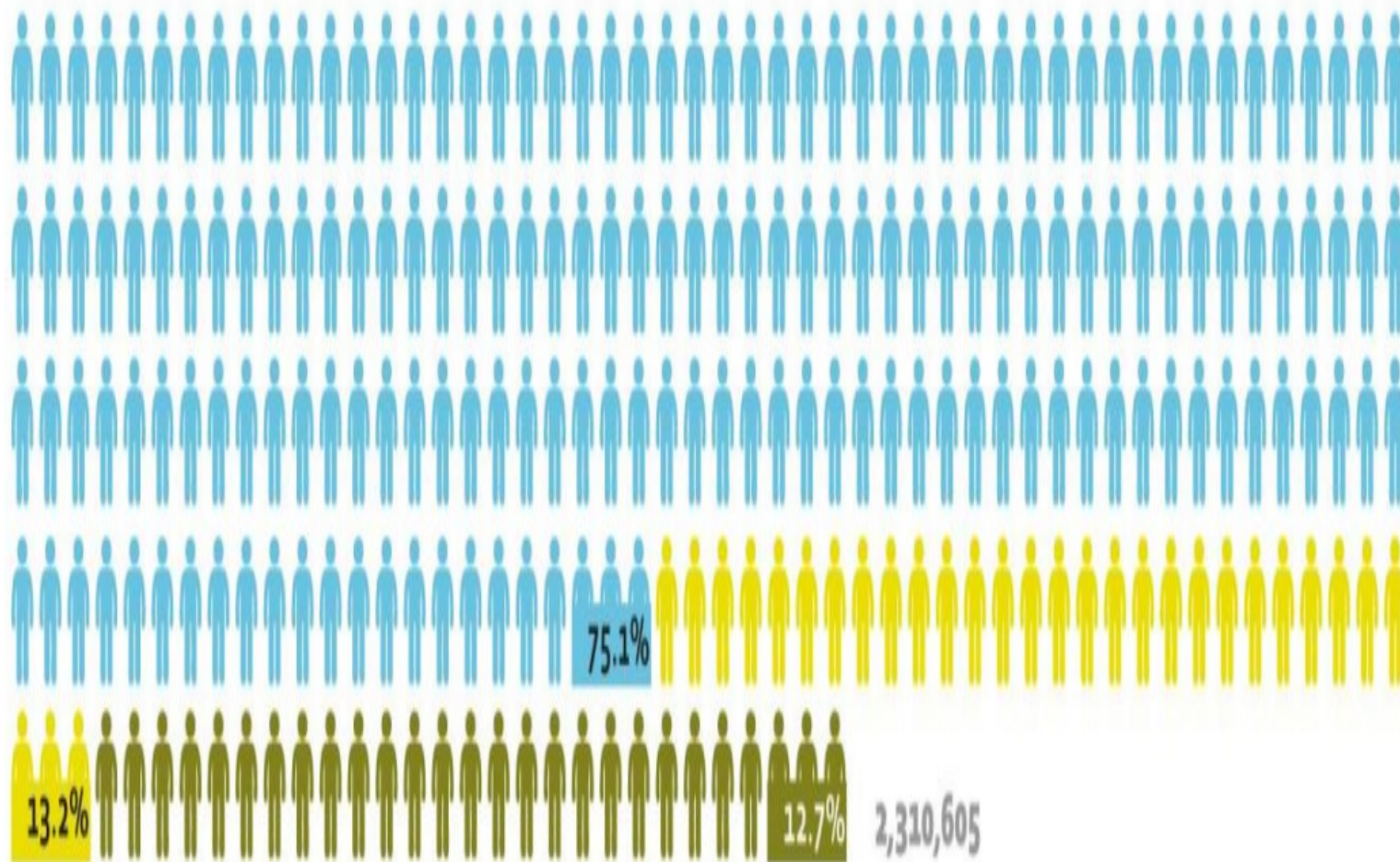
of which

2,824,984

pharmacists
are **actively practising**



Where do pharmacists work?



COMMUNITY
PHARMACY



HOSPITAL
PHARMACY



OTHER
AREAS



 = 10,000 pharmacists

Income level of pharmacist

Low income

14,079



39.2% 29.2% 31.6%

Lower-middle income

793,773



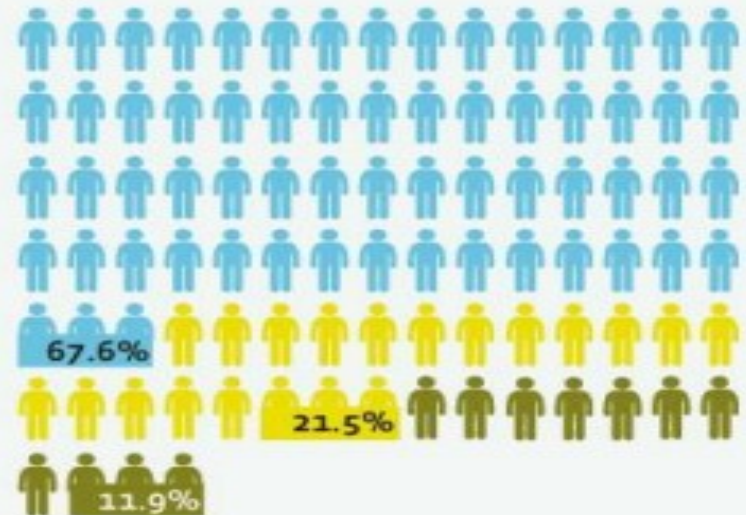
Upper-middle income

561,976



High income

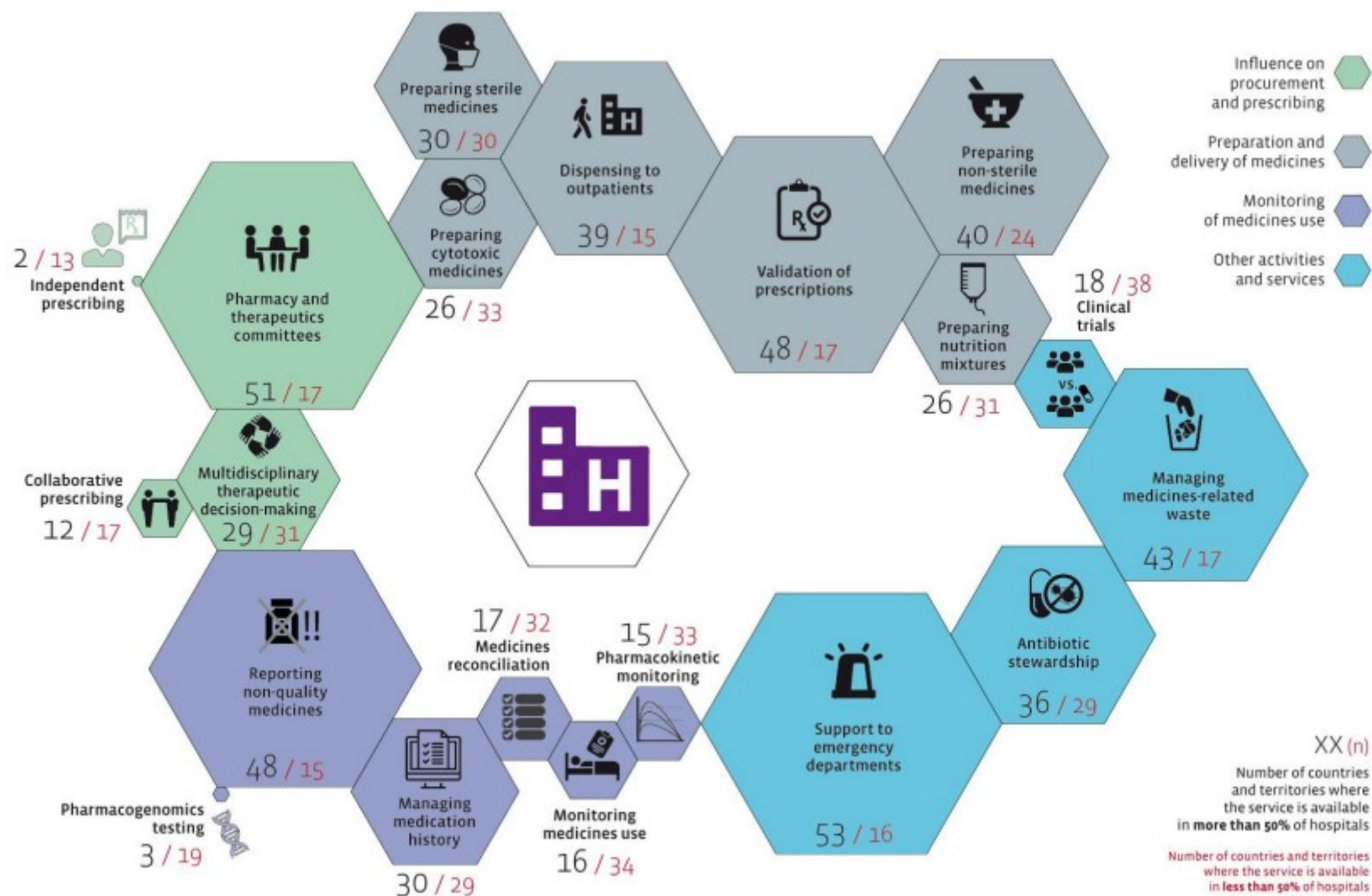
940,777



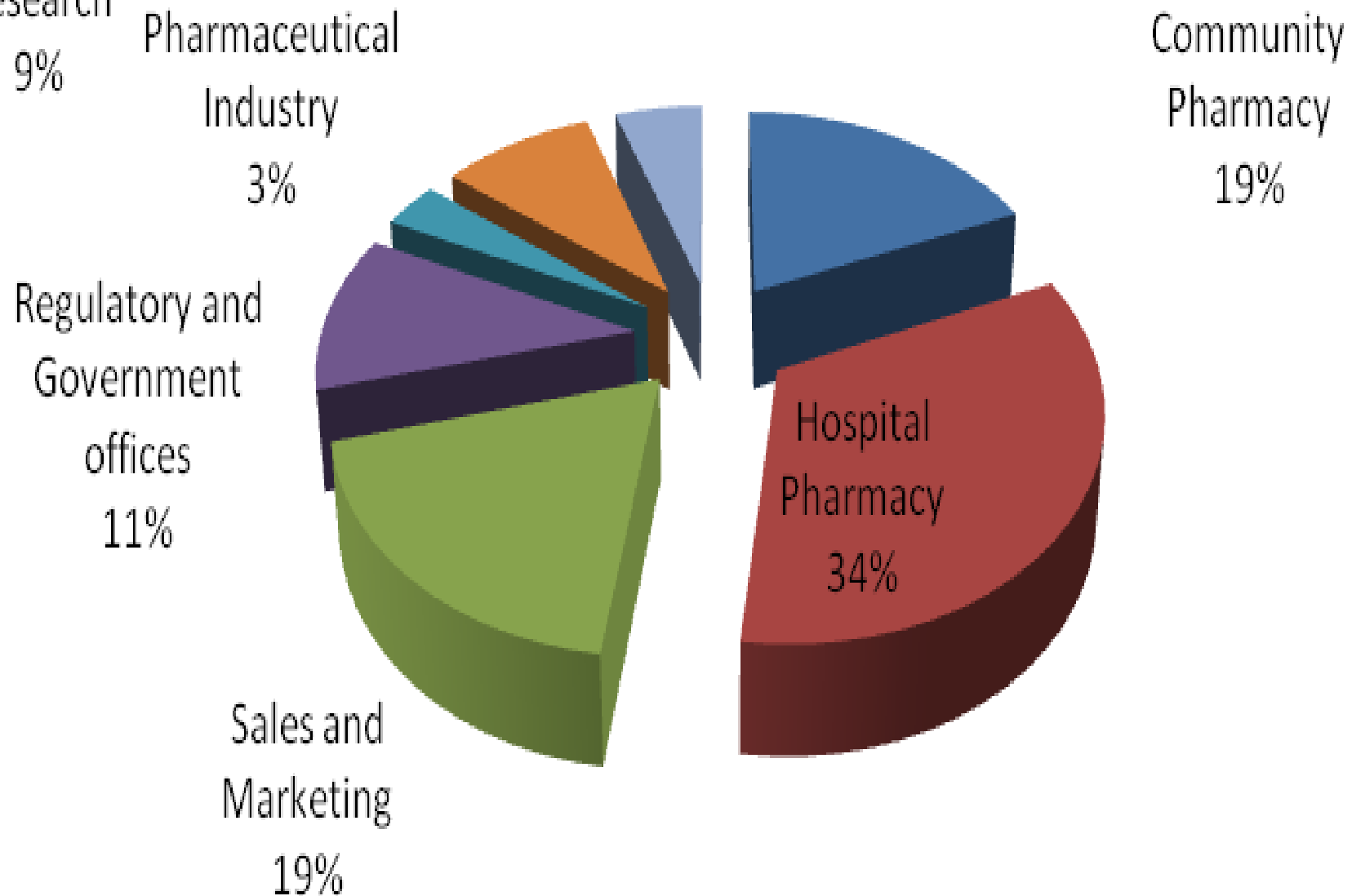
Services provided by community pharmacy



Services provided by hospital pharmacy



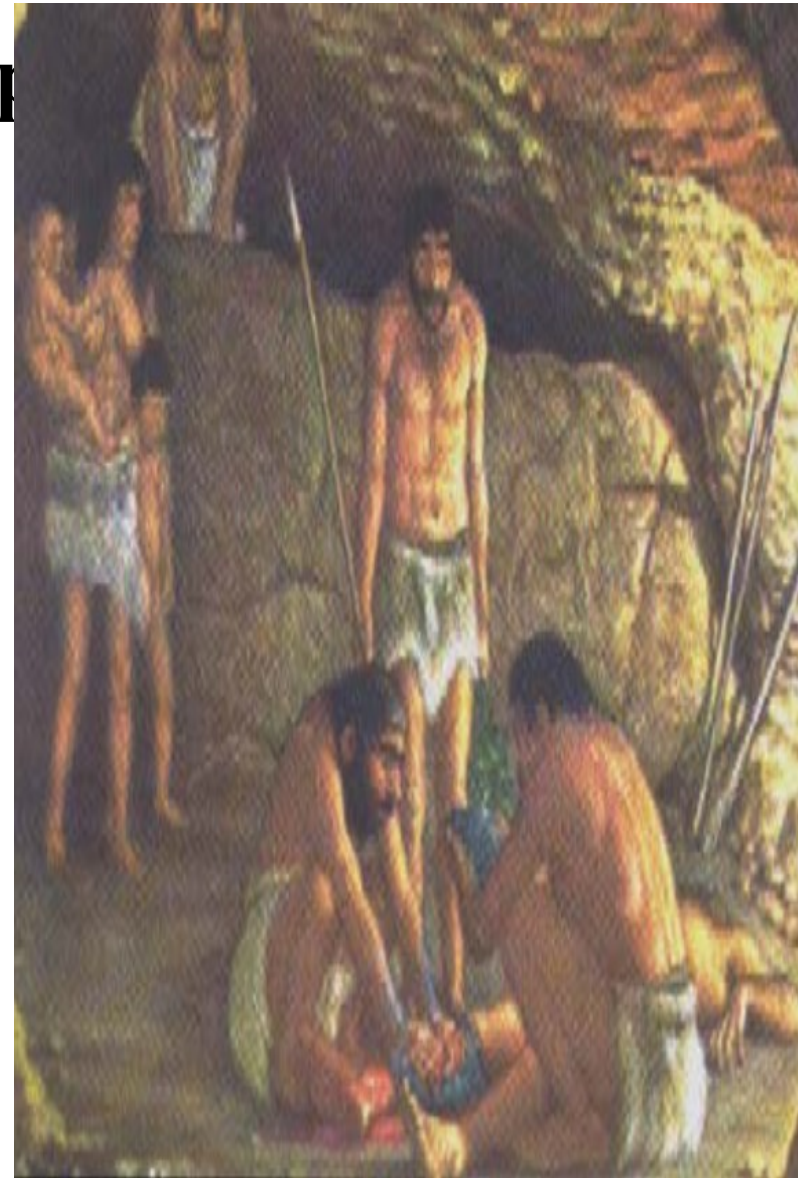
Areas of Pharmacy Practice



1.3. Evolution of p

The dawn of pharmacy, Antiquity to 50 BC

- ▶ Scientists estimate that the **Earth** began about 4.6 billion years ago, with the first **forms of life appearing as microbes** about 4 billion years ago.
- ▶ The first human ancestor *Ardipithecus ramidus* or “Ardi” appeared about 4.4 million years ago in Eastern Africa, according to the archaeological findings of Dr. Tim White and his team of researchers.



Prehistory pharmacy cont'd

- ▶ This means that **microbes** and **some of the diseases** they may cause **appeared long before humans** did.
- ▶ **Disease** is by far older than **humankind**.
- ▶ The great challenge for researchers today who study the Prehistoric Era is the lack of a written record, so we must rely on other types of evidence that **fossils and artifacts** leave us.
- ▶ Anthropologists add to our understanding of prehistoric people by studying so called “**primitive**” societies that still exist in **remote areas of the world**.
- ▶ By observing “**primitive**” **societies** anthropologists have a living window into the past

Prehistory pharmacy cont'd

- ▶ Some religious and/or medical rituals that are still practiced have remained largely unchanged for millennia, such as the pilgrimages made by very ill people high into the lakes of the Andes Mountains in Peru, led by modern day shamans who appear timeless.
- ▶ As early as 80,000 years ago, prehistoric people engraved drawings of plants and plant parts on bones and deer antlers, perhaps as a way of passing down their intimate knowledge of their environment to their descendants.

Prehistory pharmacy cont'd

- ▶ **Pharmacy** is amongst the **oldest of professions**
- ▶ Ancient man learned from **instinct**, from **observation** of birds and **beasts**.
- ▶ Pre historic people gathered **plants** for medicinal purpose
- ▶ Healing was by **trial** and **error**
- ▶ *For example, when someone is stung by a **bee**, a natural reaction is to rub the afflicted area with something **cool** and **grainy** such as mud, which **brings relief from the pain** and also has the **potential to help remove the source of the pain**, the bee's stinger*

Prehistory pharmacy cont'd

- ▶ Further, people observed what animals **ingested** and also learned by trial and error what was **helpful** in healing and what was not.
- ▶ Over time, individuals in a tribe emerged who had **special talents for healing others**, known as “**shamans**,” “**witch doctors**,” or “**medicine men**.”
- ▶ The term “**shaman**” derives from the **Siberian language** meaning “**one who knows**.”
- ▶ Thus, the role of the learned and trusted healer in the person of the shaman evolved over time into the modern **physician, pharmacist, nurse, and other health care professionals**.

Prehistory pharmacy cont'd

- Shamans developed theories of illness based on supernatural causation.
- Causes of illness might include a person who broke a tribal taboo, dishonored a god, or had a spell cast against them by another shaman.
- In order to heal the afflicted person, the patient would “confess” (today we use the term “present”) their symptoms to the shaman and the shaman would assess the problem and offer a cure.
- Even with all of our modern technology, 75% of all primary diagnoses derive from the patient’s presentation of symptoms.

Prehistory pharmacy cont'd

- ▶ It is important to note that the **shaman** treated the “whole” patient. The **mind, body, and spirit** were treated as one entity and not compartmentalized as they often are in modern health care.
- ▶ The goal of the shaman was to **cast out the “evil” or purify the patient** by means of both oral and manual rites.
- ▶ **Oral rites** were symbolic and palliative in nature and included **chants, incantations, and dances**
- ▶ **Manual rites** were physical actions performed on the patient by the shaman and included, performing a rudimentary **surgical procedure**, sucking out **poison from a wound**, or administering **herbs** for a patient to smell or swallow.

Prehistory pharmacy cont'd

- ▶ Indeed, administering herbs or potions might well have been the first form of drug therapy.
- ▶ If the illness is severe, the shaman might administer an especially harsh potion, disgusting concoction or fumigation to ward off evil.
- ▶ Shamans performed **brain surgery** known as **trepanation** to relieve pressure on the skull caused by head trauma. They performed this to **release the evil spirits** they believed were inside their patients' heads **causing pain**.

Prehistory pharmacy cont'd

- Using sharpened stone tools, the shaman made the incision on the top of the patient's skull in order to **let the demons escape**.
 - A piece of bone was cut out and would be given to the patient after the trepanation to wear as an **amulet to ward** off future demons from attacking.
-
- Trepanned Skull—
 - More than 75% patients survive after surgery.
 - Shamans also knew how to set bone fractures, perform cauterizations, and perform amputation of limbs



Pharmacy during Neolithic

- ▶ **Neolithic Age**
- ▶ About 8,000 years ago, people learned how to farm and domesticate animals, which allowed them to settle in higher density populations in more permanent settlements.
- ▶ With higher density population came political hierarchy, division of labor, and specialization that contributed to the growth of shamans who had the time and means to expand their knowledge.
- ▶ It also brought about some unintended consequences.
- ▶ many people living in close proximity to many animals created an environment for **cross-species transmission** of infectious diseases.
 - ▶ **smallpox** and mad **cow** disease
 - ▶ humans brought **tuberculosis** to cattle.
 - ▶ **Pigs** and **fowl** brought **influenzas** of varying types and degrees.
 - ▶ Dogs brought us **measles**.
 - ▶ Dogs, cats, chickens, mice, rats, and reptiles brought us **salmonella**.
 - ▶ polio, typhoid, viral hepatitis, whooping cough, and diphtheria

- Today, about 25 percent of all prescription drugs are derived from trees, shrubs, or herbs and the other 25% are derived from microorganisms.
- Astonishingly, less than 10 percent of the world's plants have been analyzed and studied for their possible therapeutic properties.

▶ **Mortar and Pestle**

- ▶ The mortar and pestle are perhaps the oldest known tools associated with early pharmaceutical work that have come to be recognized together as the icon for pharmacy itself.
- ▶ Mortars and pestles were tools used to grind crude herbs into powder.
- ▶ The modern people rightfully might be asking, did this treatment work?
- ▶ Modern medicine tells us that most physical human illness is self-limiting, so most of the time the patients recovered regardless of what the shaman did.
- ▶ Still, modern medicine is becoming more intrigued with the power of the “placebo” effect. The word “placebo” derives from Latin *placere* and literally means “I shall please.”

- Studies have repeatedly shown that if a patient has confidence in the treatment and the person administering it, the chances for recovery increase.

- Art of **primitive pharmacy** were mastered by all who practiced **domestic** medicine in the **household**.
- All forms of ancient medicine belief in supernatural orientation or magic.

Evolution.....cont'd

- Belief in magic has influenced and shaped human behavior more deeply and extensively than scientific or rationalist modes of thought.
- Magic plays an important role in many cultures
 - it provides answers to questions that cannot be answered by **existing logical or rational knowledge.**

Evolution.....cont'd

- ▶ The primary difference between a **prayer** and a **spell**
 - ▶ A **magical** practices that is correctly performed, **must** bring about the desired reaction.
 - ▶ A **prayer** is an appeal for aid from a supernatural being who has the power to grant or deny the request.
- ▶ Disease and misfortune are attributed to supernatural agents

- ▶ Magic is essential to the prevention, diagnosis, and treatment of disease.
 - ▶ ghosts, spirits, gods, sorcery, witchcraft, or the loss of one of the victim's special souls.
- ▶ Those who have the power to control the supernatural agents of disease:
 - ▶ the shaman, medicine man, wise woman, diviner, priest, soul-catcher, or sorcerer.
- ▶ Healers (Shamans) perceive **illness** is in **supernatural** terms (good and evil spirit)

Evolution.....cont'd

- ▶ Treatment (medicine) worked through supernatural means
 - ▶ Drugs as both simple tool & special substances with supernatural power
- ▶ Healers and herbalists
 - ▶ who dispense ordinary remedies
- ▶ Shamans or priest-like healers
 - ▶ who can intercede with the spirits that affect weather, harvests, hunting, warfare, conception, childbirth, disease, and misfortune.
 - ▶ performs magical acts
 - ▶ the cause of the disorder is more significant than the symptoms
 - ▶ determines the manner of treatment, be it herbs or exorcisms

Evolution.....cont'd

- Diagnostic aids
 - spirit medium, crystal gazing, and divination.
- Healer might conduct a complex ritual
 - involving magic spells
 - Incantations
 - the extraction of visible or invisible objects
 - the capture and return of the patient's lost soul.

- Examples

- patients urinate on the ground near an ant hill

- Diabetes

- Ritual scarification

- signify caste, adulthood, or the “medicine marks”
thought to provide immunization

Evolution.....cont'd

- To drive out or confuse evil spirits, the shaman may give the patient
 - A special disguise or **a new name**
 - offer attractive substitute targets
 - prescribe noxious medicines to transform the patient into an undesirable host.

Evolution.....cont'd

▶ Sumerians

- ▶ Cuneiform writing on clay tablets (2700 BC)
 - ▶ Drugs of **animal**, **vegetable** and **mineral** origin that were used in the management of diseases
 - ▶ **prescriptions** with details of the **ingredients** used in their compounding.
- ▶ Medical care was carried out by **magical healer (ashipu)** and **empirical healer (asu)**.
- ▶ Ashipu rely on **spells** and **magical stones**
- ▶ **Asu** rely on **drugs** prepared on several dosage forms
 - ▶ suppositories, pills, enemas, ointment
- ▶ The ill often went back and forth looking for cure

Apothecary Evolution.....cont'd

▶ Egyptians

- ▶ The Ebers Papyrus, 1550 BC (named after George Ebers, who purchased it in the 19th century)
 - ▶ describes prescriptions and modes of administration of drugs
 - ▶ **laxative, enemas, gargles**, inhalations, suppositories, ointments and lotions.
 - ▶ Many of the drugs listed were included in the Sumerian documents.
 - ▶ Suggested recipes usually began with a **prayer/incantation**
- ▶ Use of medicines was carried out by **priests**.

Evolution.....cont'd

▶ India

- ▶ Ayurvedic medicine was first described around 800 BC.
- ▶ Use of **drugs** together with **charms** for expelling demons and make reference to the god of medicine, **Dhanvantari**.
- ▶ The **Charaka Samhita** includes reference to drugs of **animal**, **plant** and **mineral** origin used until the first century AD.

▶ China

- ▶ Chinese pharmacy traces its origins to the emperor **Shen Nung** in about 2000 BC.
- ▶ He investigated the medicinal value of several hundred **herbs**, and wrote the first **Pen T'sao** or **native herbal**, containing 365 drugs.

Evolution.....cont'd

► Greek

- **Mixed concept** of drug or **pharmakon**(spiritual (magic) + empirical)
- Early Greek physician diagnosed the **natural causes for illness**, while still **not rejecting the use of supernatural** healing in conjunction with empirical remedies.
- Some people beset with persistent afflictions traveled to a temple of the god **Asklepios**, where they would sleep with the hope of being visited during the night by the god (of Medicine and Healing) or his daughter **Hygeia**, who carried a magical serpent and a bowl of healing medicine.

Asclepius and his daughter Hygieia

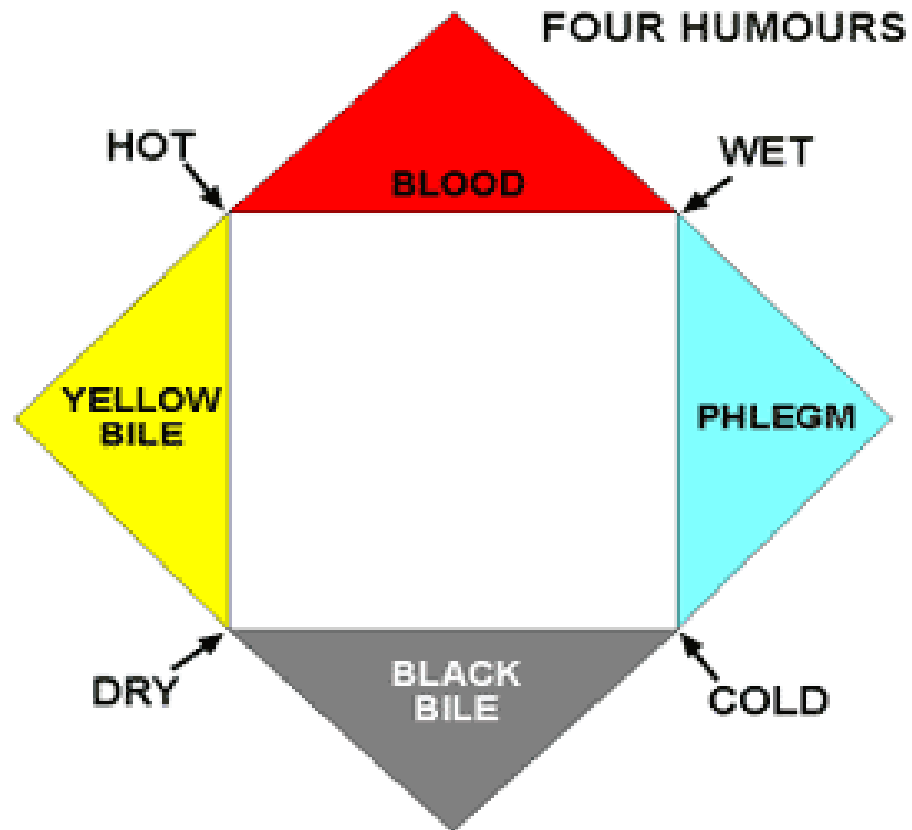


Evolution.....cont'd

▶ Hippocrates

- ▶ Considered to be the **father** of medicine.
- ▶ His works placed emphasis on treating the patient with **minimal reference to magical and religious powers.**
- ▶ Constructed **rational** explanation of illness
- ▶ Conceptual link between **environment** and **humanity** (humors of body)
 - ▶ Earth, air, fire and water
 - ▶ Black bile, blood, yellow bile and phlegm

Four humour



- **Dioscorides**

- Prepared the document **De Materia Medica** around AD 60-78.

- gives details about medicinal herbs including side-effects associated with their administration.

- Became the **standard encyclopedia of drugs**

Materia medica



Evolution.....cont'd

- **Galen**
 - A Greek physician who practised in Rome around 160 AD.
 - He compiled medical knowledge of the time drawing on the documents by **Hippocrates** and **Dioscorides**.
 - The **humoral** system of medicine gained **ascendancy**
 - Devised an elaborate system to balance that **attempted to balance the humor** of an ill individual by using **drugs of supposedly contrary nature**

Evolution.....cont'd

- ▶ E.g. Applying cucumber, a cool and wet drug to treating external inflammation
- ▶ Bleeding to remove the excess blood that cause illness
- ▶ He also advocated the use of polypharmaceutical preparation ”**shotgun prescription**”
- ▶ He described the use of formulations made up of numerous plants which were referred to as ‘**galenicals**’.

Middle ages

- ▶ **Middle age (400AD to 1453)**
 - ▶ Fall of **Rome** to fall of **Constantinople**
 - ▶ The **flourishing of Islamic civilization** in pharmacy
 - ▶ In the Arab world, a large number of texts including documents related to medicine and works by **Galen** were **translated** into Arabic
 - ▶ Islamic medical men like **Rhazes** (860-932) and **Avicenna** (980-1063) **added** to the writings of the Greeks.
 - ▶ **Rejected** the old idea that **foul tasting** medicines worked best.
 - ▶ DFs become **elegant** and **palatable** through silvering and gilding of **pills** and the use of **syrups**.

The separation of pharmacy from Medicine

- Pharmacy first separated from medicine by the **Arabs** in Baghdad in the 8th century
- It was an integral part of a deliberate public health program in which **apothecaries and physicians** played separate and distinct roles.
- In **Europe** pharmacy first became legally separated from medicine in **1231** AD in Sicily and southern Italy by **Frederick** II of Hohenstaufen
- Frederick's edict contained three decrees:
 - the pharmaceutical profession was to be **separated** from the medical profession
 - the pharmaceutical profession should be **supervised officially**
 - pharmacists should take an **oath** to prepare drugs reliably, according to skilled art, and in a uniform suitable quality

The Renaissance and early modern Europe

- ▶ It was time for **new ideas**
 - ▶ Reinterpretation of the old classical themes
 - ▶ Exploration in laboratory
- ▶ **Johan Gutenberg** (1453) started **printing** with movable type (starting an information revolution)
 - ▶ has profound effect on the **study of plant** drugs
 - ▶ Easy to reproduce **illustration** of the plants
- ▶ **Valerius Cordus** (1515 - 1544) wrote a book called “***Dispensatorium***” in 1546 which became **the official standard** for the preparation of medicines in the city of Nuremeberg and generally considered the **first pharmacopeia**.

The Renaissance.....cont'd

- ▶ **Paracelsus** (1493-1541) introduced two ideas:
 - 1) disease might be localized **in a specific organ** and such conditions could be treated internally using the **chemical** properties of medicinal agents and
 - 2) some **plants and other substances** contained minute quantities of **active chemicals**, which could be removed by making **tinctures, extracts, and essences**.
- ▶ He was advocator of **chemically prepared** drugs from crude plants and mineral substances
- ▶ He was also preaching “***the doctrine of signature***”

The Renaissance.....cont'd

- ▶ A belief that **God** had placed a **sign** on healing **substances** indicating their use against **disease**
 - ▶ E.g. liverwort resembles a liver, so it must be good for liver ailments
 - ▶ E.g2, a plant remedy for jaundice might sport yellow flowers a
 - ▶ E.g 3 remedies for heart disease might be found in plants with heart-shaped leaves.
- ▶ In 17th, 18th and into the beginning of 19th century as chemistry emerged
- ▶ small minority of practicing pharmacist made significant investigation into the ***chemistry of drugs and*** isolated many drugs that are still use today and contributed much to the general chemical knowledge

The Renaissance.....cont'd

- **Tobacco, guaiac, cascara sagrada, ipecac and cinchona** bark were identified new plant drugs from the new world.
- In 1820, **Quinine** was extracted from cinchona bark

Modern Pharmacy (17th and 18th century)

- First medicinal patent-for Epsom salts (magnesium sulfate compound) (1698) as a healing agent and pain reliever
- Three trends:
 - Greatly **simplified formulas**
 - Increasing **use of chemicals**
 - Increased use of **Alcohol** as a solvent
- Development of the hypodermic needle (1853)

Background to Modern Pharmacy cont...

Immunology and Medical Bacteriology

- The first disease against which an artificial immunity was produced was **smallpox**
- **Edward Jenner** (1749-1823) credited for making small pox **vaccination** safe
- The germ theory of disease - French chemist Louis Pasteur (1822-1897) and the German physician Robert Koch (1843-1910)
- British surgeon Joseph Lister introduced **antisepsis** in the 1860's
- This development had its impacts on:
 - making biological products one of the responsibility of the pharmacist.
 - Making the manufacture and use of sterile products possible

Background to Modern Pharmacy cont...

Chemotherapy

- Paul Ehrlich (1854-1915) was the founder of this new concept.
- Ehrlich and coworkers found **Arsphenamine** (trade named “Salvarsan”) effective in treating **sleeping sickness** and also **syphilis**.
- Chemotherapy –
“the direct attack on morbid agents in the body by means of chemicals, without doing unbearable harm to the cells containing the microbes or to the body at large”
- Prontosil (a pro-drug for sulfanilamide) was announced by Gerhard Domagk in 1935 to be effective against streptococcal infections.

Background to Modern Pharmacy cont...

▶ Vitamins and Hormones

- ▶ The term vitamin was first introduced in 1912
- ▶ The first hormone to be isolated was **adrenaline** in 1901.
- ▶ **Insulin** was discovered by F.G. Banting, in 1922.
- ▶ The first clinical trials of oral contraceptives 1956 by Gregory Pincus

▶ Psychopharmacology

- ▶ **chlorpromazine** (1952) was the first striking therapeutic success in this area

The development in the basic sciences was the basis for the increased discovery of new and effective therapeutic agents

Assignment 2

- ▶ Explain the origins of humoral theory and how it affected the development of pharmacy.
- ▶ Explain how the Romans built upon Greek medical and pharmacy achievements.
- ▶ Explain what a patient experienced at a Temple of Asclepius.
- ▶ Explain the Ayurvedic theory of illness.
- ▶ Describe the Chinese theory of illness.

1.4. History of Drugs and Dosage Forms

1.4.1. Introduction

- **Drug** defn
 - “a substance intended for use in the **diagnosis**, **cure**, **mitigation**, treatment, or prevention of disease in humans or animals”.
- **Dosage form** is the means by which drug molecules are delivered to sites of action within the body.
- Exact time man started using Drugs and DFs is not known.

Types of dosage forms:

The need for dosage forms:

- 1- Accurate **dose**.
- 2- **Protection** e.g. coated tablets, sealed ampules.
- 3- Protection from **gastric juice**.
- 4- Masking **taste** and **odour**.
- 5- **Placement** of drugs within body **tissues**.
- 6- **Sustained** release medication.
- 7- **Controlled** release medication.
- 8- Optimal drug action.
- 9- **Insertion** of drugs into body cavities (rectal, vaginal)

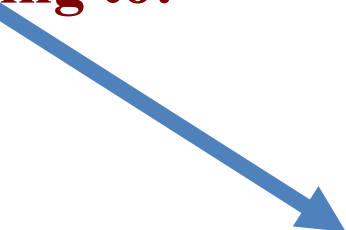
Types of dosage forms (Cont.):

They are classified according to:



Route of administration

- ▶ Oral
- ▶ Topical
- ▶ Rectal
- ▶ Parenteral
- ▶ Vaginal
- ▶ Inhaled
- ▶ Ophthalmic
- ▶ Otic



Physical form

Solid
Semisolid
liquid

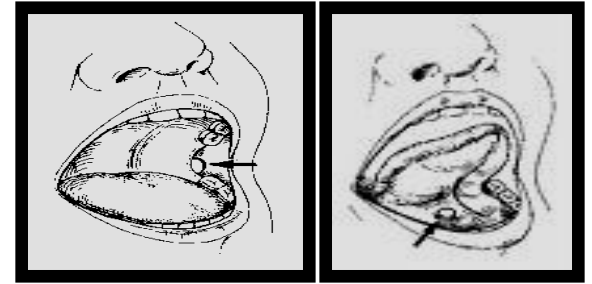
Oral dosage forms:



1-Tablet:

- ▶ A **tablet** is a hard, compressed medication in round, oval or square shape.
- ▶ The excipients include:
 - ***Binders, glidants*** (flow aids) and **lubricants** to ensure efficient tableting.
 - ***Disintegrants*** to ensure that the tablet breaks up in the digestive tract.
 - ***Sweeteners*** or flavours to mask the taste of bad-tasting active ingredients.
 - **Colourant** to make uncoated tablets visually attractive.

2-Buccal and sublingual tablet:



- **Sublingual** and **buccal** medications are administered by placing them in the mouth, either under the tongue (**sublingual**) or between the gum and the cheek (**buccal**).
- The medications dissolve rapidly and are absorbed through the mucous membranes of the mouth, where they enter into the bloodstream.
- Avoid the acid and enzymatic environment of the stomach and the drug metabolizing enzymes of the liver.
- Examples of drugs administered by this route: e.g. vasodilators, steroidal hormones.

3-Effervescent tablet:



- ▶ **Effervescent tablets** are uncoated tablets that generally contain **acid substances** (citric and tartaric acids) and **carbonates or bicarbonates** and which react rapidly in the presence of water by releasing **carbon dioxide**.
- ▶ They are intended to be dissolved or dispersed in water before use providing:
 - ▶ Very **rapid** tablet dispersion and dissolution.
 - ▶ pleasant tasting carbonated drink.

4- Chewable tablet:



- They are tablets that chewed prior to swallowing.
- They are designed for administration to children e.g. vitamin products.

Hard gelatin capsule

Soft gelatin capsule

5- Capsule:



- **A capsule** is a medication in a **gelatin container**.
 - Advantage: mask the unpleasant taste of its contents.
- The two main types of capsules are:
 - 1- hard-shelled capsules**, which are normally used for dry, powdered ingredients,
 - 2- soft-shelled capsules**, primarily used for oils and for active ingredients that are dissolved or suspended in oil.

6- Lozenge:



- It is a solid preparation consisting of **sugar and gum**, the latter giving strength and cohesiveness to the lozenge and facilitating slow release of the medicament.
- It is used to medicate the **mouth** and **throat** for the slow administration of indigestion or cough remedies.

8- Dental Cones:



- A tablet form intended to be placed in the empty socket following a tooth extraction, for preventing the local multiplication of pathogenic bacteria associated with tooth extractions.
- The cones may contain an antibiotic or antiseptic.

10- Granules:



- They are consisting of solid, dry aggregates of powder particles often supplied in single-dose sachets.
- Some granules are placed on the tongue and swallowed with water, others are intended to be dissolved in water before taking.
- **Effervescent granules** evolve carbon dioxide when added to water.

11- Powder (Oral):



There are two kinds of powder intended for **internal** use.

- ▶ *Bulk Powders* are multidose preparations consisting of solid, loose, dry particles of varying degrees of fineness.
 - ▶ They contain one or more active ingredients, with or without excipients and, if necessary, coloring matter and flavoring substances.
 - ▶ usually contain **non-potent** medicaments such as antacids
- ▶ *Divided Powders* are single-dose presentations of powder (for example, a small sachet) that are intended to be issued to the patient as such, to be taken in or with water.

12- Powders for mixtures:

- The mixed powders may be stored in dry form and mixture prepared by the pharmacist when required for dispensing , by suspending the powders in the appropriate vehicle.

13-Liquid preparations:

a- Oral solution:

- ▶ Oral solutions are clear liquid preparations for oral use containing **one or more active ingredients dissolved** in a suitable vehicle.



b- Oral emulsion:

- ▶ Oral emulsions are stabilized **oil-in-water** dispersions, either or both phases of which may contain dissolved solids.



c-Oral suspension:



- Oral suspensions are liquid preparations for oral use containing one or more active ingredients **suspended** in a suitable vehicle.
- Oral suspensions may show **a sediment** which is readily **dispersed on shaking** to give a uniform suspension which remains sufficiently stable to enable the correct dose to be delivered.

13-Liquid preparations (Cont.):

d- Syrup:

- It is a **concentrated aqueous solution** of a **sugar**, usually sucrose.
- **Flavored** syrups are a convenient form of **masking** disagreeable tastes.

e- Elixir:

- It is pleasantly flavored clear liquid oral preparation of potent or nauseous drugs.
- The vehicle may contain a **high proportion of ethanol** or **sucrose** together **with antimicrobial preservatives** which confers the stability of the preparation.

13-Liquid preparations (Cont.):

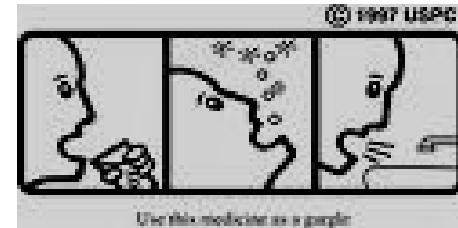
f- Linctuses:

- Linctuses are **viscous**, liquid oral preparations that are usually prescribed for the relief of **cough**.
- They usually contain a **high proportion of syrup** and **glycerol** which have a **demulcent** effect on the membranes of the throat.
- The dose volume is small (5ml) and, to prolong the **demulcent** action, they should be taken **undiluted**.

13-Liquid preparations (Cont.):

h- Gargles:

- They are **aqueous solutions** used in the prevention or treatment of **throat** infections.
- Usually they are prepared in a **concentrated** solution with directions for the patient to **dilute** with warm water before use.



i- Mouthwashes:

These are similar to **gargles** but are used for **oral hygiene** and to treat infections of the **mouth**.



Topical dosage forms:

1- Ointments:

- Ointments are **semi-solid**, **greasy** preparations for application to the **skin**, **rectum** or **nasal mucosa**.
- The **base** is usually **anhydrous** and **immiscible** with skin secretions.
- Ointments may be used as **emollients** or to apply **suspended or dissolved medicaments** to the skin.



Topical dosage forms (Cont.):



2- Creams:

- Creams are **semi-solid** emulsions, that is mixtures of **oil** and **water**.
- They are divided into two types:
 - A- oil-in-water (O/W) creams:** which are composed of **small** droplets of **oil** dispersed in a continuous **aqueous** phase.
 - Oil-in-water creams are **more comfortable** and **cosmetically** acceptable as they are **less greasy** and more **easily washed off** using water.

Topical dosage forms (Cont.):

B- water-in-oil (W/O) creams:

- which are composed of **small** droplets of **water** dispersed in a continuous **oily** phase.
- Water-in-oil creams are also more moisturising as they provide an **oily** barrier which **reduces** water loss from the stratum corneum, the outermost layer of the skin.

Topical dosage forms (Cont.):



▶ 3- Gels (Jellies):

- ▶ Gels are semisolid system in which a **liquid** phase is constrained within a 3-D polymeric matrix (consisting of natural or synthetic gum) having a high degree of physical or chemical cross-linking.
- ▶ They are used for **medication**, **lubrication** and some miscellaneous applications like carrier for **spermicidal** agents to be used intra vaginally .



Topical dosage forms (Cont.):

4- Poultice:

- It is soft, viscous, pasty preparation for external use. They are applied to skin while they are hot. Poultice must retain heat for a considerable time because they are intended to supply warmth to inflamed parts of body.
- E.g. Kaolin poultice (B.P.C.)



Topical dosage forms (Cont.):

5- Pastes :

- ▶ Pastes are basically ointments into which a high percentage of insoluble solid has been added
- ▶ The extraordinary amount of particulate matter stiffens the system.
- ▶ Pastes are less penetrating and less macerating and less heating than ointment.
- ▶ Pastes make particularly good protective barrier when placed on the skin, the solid they contain can absorb and thereby neutralize certain noxious chemicals before they ever reach the skin.

—

Topical dosage forms (Cont.):

- ▶ Like ointments, paste forms an unbroken relatively water – impermeable film unlike ointments the film is opaque and therefore can be used as an effective sun block accordingly.
- ▶ Pastes are less greasy because of the absorption of the fluid hydrocarbon fraction to the particulates.

Topical dosage forms (Cont.):

6- Dusting powders:

- These are free flowing very fine powders for external use.
- Not for use on open wounds unless the powders are sterilized.



Topical dosage forms (Cont.)



- **7- Transdermal patch:**
 - A **transdermal patch** or **skin patch** is a medicated adhesive patch that is placed on the skin to deliver a **specific** dose of medication through the **skin** and into the **bloodstream**.
 - An advantage of a transdermal drug delivery route over other types such as oral, topical, etc is that it provides a controlled release of the medicament into the patient.
 - E.g scopolamine for motion sickness.

Topical dosage forms (Cont.):

8-Plasters:

- Plasters are solid or semisolid masses adhere to the skin when spread upon **cotton felt line or muslin** as a backing material and they are mainly used to,

A- Afford protection and mechanical support.

B- Furnish an occlusive and macerating action.

C- Bring medication into close contact with the surface of the skin.



Topical dosage forms (Cont.):

9- Liniments:

- Liniments are fluid, semi-fluid or, occasionally, semi-solid preparations intended for application to the skin.
- They may be alcoholic or oily solutions or emulsions.
- Most are massaged into the skin (e.g. counter-irritant).
- Liniments should not be applied to broken skin.



Topical dosage forms (Cont.):

10-Lotions:

- These are fluid preparations (aqueous) for external application without friction.
- They are either dabbed on the skin or applied on a suitable dressing and covered with a waterproof dressing to reduce evaporation.



Topical dosage forms (Cont.):



11. Collodion is a solution of nitrocellulose in ether or acetone, sometimes with the addition of alcohols.

- Its generic name is pyroxylin solution.
- It is highly flammable.
 - ▶ As the solvent evaporates, it dries to a celluloid-like film.
 - ▶ Compound Wart Remover consists of acetic acid and salicylic acid in an acetone collodion base used in Treatment of warts by keratolysis.

Topical dosage forms (Cont.):

12- Paints:

- Paints are liquids for application to the skin or mucous membranes.
- Skin paints contain **volatile** solvent that evaporates quickly to leave a **dry resinous film** of medicament.
- **Throat paints** are more **viscous** due to a high content of glycerol, designed to prolong contact of the medicament with the affected site.



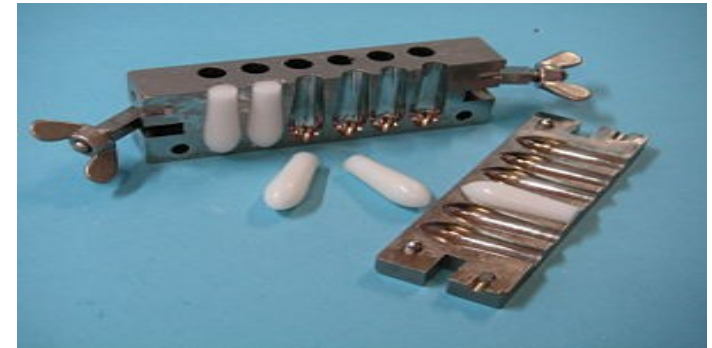
Rectal dosage forms:

► Suppository

- It is a small solid medicated mass, usually **cone-shaped**, that is inserted either into the rectum where it **melts** at body temperature.

► Enema

- An **enema** is the procedure of introducing **liquids** into the rectum and colon via the anus.
- Evacuant enema: constipation
- Retention enema: local vs systemic



Vaginal dosage forms:

- **Pessaries**

- are solid medicated preparations designed for insertion into the vagina where they **melt** or **dissolve**.



- **Douche**

- is a device used to introduce a stream of water into the body for **medical or hygienic** reasons



1.5. Pharmacy in Ethiopia: development and types of pharmacy settings

- ▶ **Religious** practices play a large part in the healing process for Ethiopians.
- ▶ The **skill** of traditional health practitioners is '**given by God**'
- ▶ **knowledge** on traditional medicines is passed **orally**
 - ▶ father to a **favorite** child, usually a **son**
 - ▶ acquired by some spiritual procedures.
- ▶ Traditional Healing knowledge is guarded by **certain families or social groups**
- ▶ Healing, protection and promotion of human physical, spiritual, social, mental and material wellbeing

Pharmacy in Ethiopia.....cont'd

▶ *Traditional medicine practitioners and practices*

- ▶ Bonesetters '*Wogesha*',
- ▶ Traditional birth attendants “*yelimd awalaj*'
- ▶ Tooth extractors
- ▶ Herbalists
 - ▶ 'debtera',
 - ▶ 'tenquay' (witch doctors)
- ▶ Spiritual healers
 - ▶ '*weqaby*'
 - ▶ '*kalicha*'

Pharmacy in Ethiopia.....cont'd

- ▶ Healers obtain their drugs mainly from natural substances
 - ▶ plants, animals and minerals.
- ▶ Dosage forms
 - ▶ liquids, ointments, powders and pills.
- ▶ Route of administration
 - ▶ topical, oral and respiratory
- ▶ The healers imposed restriction when certain types of drugs were taken by patients.

Pharmacy in Ethiopia.....cont'd

- **Foreign** traders, and **missionaries** brought d/t medicines in 15th C
- No evidence of the **profession** of the missionaries
- **Salinbini**, presented some drugs to Ras Alula (January 1886)
- Emperor **Minilik** - the first Ethiopian “drug importer and dispenser”
- He received drugs used for ailments such as:
 - fever/malaria, inflammation and rheumatism, intestinal worms, cough, and anti venom for snake bite
- The first proper pharmacy setup 1880 in the **palace**

Pharmacy in Ethiopia.....cont'd

- “**Pharmazie Le Jorge**” 1st retail pharmacy giving service to the general public (Nov, 1911) by Dr **Mereb** - Ras Makonnen Avenue
- **St. Jorge Pharmacy** (1914), 2nd by **Walter Zahn**
 - The pharmacy had **two** entrances, one for **natives** and one for **Europeans**.
- By early 1930's Ethiopians started to be patrons of Zahn's pharmacy

Pharmacy in Ethiopia.....cont'd

- The first legislation regulating the work of the **doctor, pharmacist, midwife and veterinarian** was promulgated in **1930** (July 18).
- The legislation gave **exclusive right** of dispensing and stocking drugs to those with formal **medical training**
- EPA was formed officially in November 1974-69 members

Pharmacy in Ethiopia.....cont'd

- ▶ Among the pharmacies established during and after 1925:
 - ▶ Pharmacie Principale established by Armenian, K. Arabian in 1925
 - ▶ Pharmacie Normale in Rue de la Gare by Armenian, N. Broussalian in 1926
 - ▶ Pharmacie Greco-Ethiopienne by Greek, G. Athanassiades in 1927
 - ▶ Pharmacie Ethiopienne (1928) by the Greek, D Zagorides (owner) and run by a pharmacist called Francis
 - ▶ Pharmacie Imperial by Armenian S. Latifian in 1932
 - ▶ Pharmacy Centrale by Hans Jacobsohn in 1935
 - ▶ Droguerie L'Ethiopie (drug store), founded in 1933 by a Greek called D. Caravidas

Types of Pharmacy Settings in Ethiopia (the Current Situation)

- ▶ **Pharmacy:** managed under the responsibility of a registered/licensed **pharmacist**
 - ▶ minimum experience of **five** years.
 - ▶ only established in a place where there is at least **one hospital giving service to the public.**
- ▶ **Drug Store/Animal Drug Store:** managed under the responsibility of a registered **druggist or veterinary doctor**
 - ▶ at least 5 years of experience
 - ▶ established in a place where there is at least **one health center/animal clinic giving service**

Types of Pharmacy Settings in Ethiopia cont...

- **Rural Drug Vendor/Animal Rural Drug Vendor:** managed under the responsibility of a registered **pharmacy technician or assistant veterinary doctor**
 - at least **3 years** of experience.
 - established in a place where there is at least **one health center or clinic/animal clinic** managed by an assistant veterinary physician
- PFSA vs FMHACA

Pharmaceutical and Medical Terminologies

Learning Objectives

At the end of the lecture students will be able to:

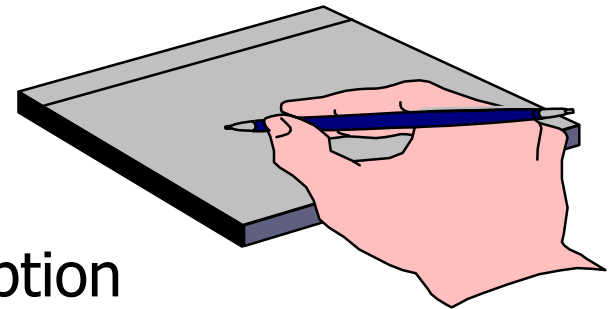
- Define and describe prescription
- List types of prescriptions and describe its importance
- Identify the major parts of prescription
- Describe the proper meaning and spelling of the pharmaceutical and medical abbreviations
- Reduce an entire phrase to a single understandable word

Prescription :

- A request from a legal prescriber for compounding or supply of medicine, dressing or surgical appliance to a patient
- Why we need Prescription??

- Prescriptions are required for all medications that :
 - require supervision/follow-up
 - addictive and carry the potential of being abused
 - cause health threats from side effects if taken incorrectly, e.g. cardiac drugs, insulin, and antibiotics

- Used as a short form for prescription only drugs (POM) to distinguish from over-the-counter drugs (OTC)
- POM: licensed medicines that are regulated by legislation to require a prescription before it can be obtained
 - Special **Vs** Conventional prescription
- OTC: drugs which don't require prescription
E.g. Paracetamol



Differences between POM and OTC medications

Prescription Only Medicines	Over-the-Counter Medicines
■ Require a written prescription ■ Authorizes to dispense particular medication	■ Can be bought without a prescription
■ Are prescribed for the treatment of a specific medical problem	■ Are intended for relief of minor ailments
■ Are usually more powerful and have more side effects than OTC medications	■ Are considered safe (API & Dose) if warnings and directions are followed... harmful if misused

NB: Different jurisdictions have different definitions of what constitutes a POM drug

Parts of the Written Prescription

- A prescription consists of four main parts

1.Superscription

2.Inscription

3.Subscription

4.Signatura

XYZ HOSPITAL
POBOX 0000000
TEL. 0110000000

Prescriber's office Info.

Patient Info. → Patient's NameSexAge.....
 Address: Region.....Town.....Woreda.....
 Kebele.....House No.....Tele..... Card No.....

Superscription → Rx:
 Hydrocortisone 0.5
 Sodium Salicylate 24.5
 Div. in caps #L

Signa → Sig: 1 cap t.i.d

Prescriber's Information → Prescriber's Name.....
 Qualification.....
 Registration No.....
 Signature.....Date.....

#21

Subscription

Or Amox 250mg tab

Inscription

1. The **superscription**:

- Consists the heading where the symbol Rx is found
 - Rx: abbreviation for **recipe**, Latin for **you take**
 - Rx symbol comes before the inscription

2. The **inscription**:

- body of the prescription
- Provides:
 - ✓ Name
 - ✓ Dose or quantities of the chief ingredients
 - ✓ Dosage form, such as tablet, suspension, capsule, syrup

3. The **subscription**

- ✓ Express quantity
- ✓ gives specific directions **for pharmacist** on how to compound the medication
- ✓ directions usually expressed in abbreviations
- ✓ E.g. Disp tabs **No c meaning "dispense 100 tablets."**

4. The **signatura**

- also called sig or transcription
- gives instructions to the patient on ***how, how much, when, and how long*** the drug is to be taken
- instructions are preceded by the symbol **"S", "Signa"** or "Sig." from the Latin, meaning, "mark thou" =
Label

Parts of Prescription Cont'd

- Directions are transcribed and labeled in to the container of dispensed medication

E.g. " Caps I t.i.d. pc," = " take one capsule three times daily after meals"

Prescription Labels

- Should include important information/instructions
- For the patient's health and well-being, the patient:
 - needs to know exactly what the directions on the medication labels mean
 - advise the patient on the best time and the best way to take medication to get the most benefit

Parts of Prescription Cont'd

- Sometimes, however, the labels can be confusing b/c instructions are not clear

Example:

- ☞ Take as directed: What were the directions?
- ☞ Take 4 times a day: Around the clock or just during waking hours?
- ☞ Take as needed: What determines need?
- ☞ Take before bedtime: Immediately before sleeping or 1-2 hours before?

- Medication orders for inpatients in hospitals and other institutions are written by other forms
- Commonly called “physician’s order sheet”
- Type of form used varies between institutions and even within the institutions depending on the unit rendering the care

Parts of Prescription Cont'd

- Before 1960s, most prescriptions were written using Latin phrases and symbols
- Thus prescribers and pharmacists needed to know Latin
- Present day prescriptions are written in English but often you still find Latin abbreviations used
 - potential to cause medication errors if not properly interpreted
- So knowing and understanding the basic Latin abbreviations is an important step in dispensing prescriptions

Use of Latin Abbreviations in the Prescription: Examples

- Sig. I tab po qid pc
 - ▶ Take one tablet by mouth 4 times a day, after meals
- Sig. instill ii gtts o.s. t.i.d.
 - ▶ Instill two drops in the left eye three times a day
- Rx: Propranolol 1mg/ml Ampoule
Sig: I amp. IV stat
- Rx: Propranolol 40 mg tab
Sig: I tab po qd for 7 days

Exercise: Translate the following abbreviations

▶ Sig. two tsp syrup t.i.d.

▶ Sig. two tbsp syrup q.d.

▶ Sig: I tab p.o q4h ac

▶ Translate the following hospital medication orders

Start IV

IV #1, NS 1000ml + 20mEq KCl @ 125ml/hr

IV #2, IV#3 NS 1000mL @ 125ml/hr

Abbreviations used to indicate Route of Administration or Administration Site***

Term	Meaning
IA	Intra-arterially, inject into artery
ID	intradermal, inject into skin
IM	intramuscular, inject into muscle
IV	intravenous, inject or infuse into vein, rate specified. For intravenous fluid therapy IV also refers to the infusion fluid.
IVP, IV bolus	intravenous push, a rapid injection into vein
KVO	keep vein open. A slow infusion. The rate is set by institution policy or specified used to maintain an intravenous catheter patent.
R, pr, rect.	Rectal
SL	sublingual, Place under tongue and allow to dissolve
subq., s.c	subcutaneously, inject below the skin
Top	topically, apply locally to affected area
vag.	Vaginally

NB: The oral route is always assumed if no route is specified (if po is possible)

Introduction to Pharmaceutical Terminologies

Phar-Terminologies Cont'd

- **Absorption:** the uptake of substances **into** or **across tissues**
 the process that occurs from the time a drug enters the body to the time it enters the bloodstream to be circulated
- **Adsorption:** the action of a substance in attracting and holding other materials or particles **on its surface**

Phar-Terminologies Cont'd

- **Adsorbent:** a drug that binds other chemicals onto its surface
 - ➡ Used to reduce the free availability of toxic chemicals
 - E.g. activated charcoal, kaolin
- **Analgesic:** an agent that relieves pain without causing loss of consciousness
 - E.g. Aspirin (ASA), Paracetamol

Phar-Terminologies Cont'd

- **Sedative:** a drug that allays irritability and excitement

E.g. Phenobarbital

- **Hypnotic :** an agent that induces sleep, e.g. Chloral hydrate
- Sedative-hypnotic drugs depress the CNS in a dose-dependent fashion:
- ➡ sedation > sleep > anesthesia> coma > fatal depression of respiration and CV regulation E.g. Barbiturates
- **Anesthetic:** an agent that produces loss of sensation with or without loss of consciousness

Phar-Terminologies Cont'd

- **Antacid:** an agent which counteracts acidity

E.g. $\text{Al}(\text{OH})_3$, $\text{Mg}(\text{OH})_2$

- **Antidote:** an agent that counteracts the effect of poison or toxin E.g.
activated charcoal

- **Antieczematic:** an agent that is effective against eczema

E.g. clobetasol propionate

redness, skin edema (swelling), itching,
cracking, oozing, or bleeding



Phar-Terminologies Cont'd

- ▶ **Antifungal:** an agent that destroys or suppresses the growth or reproduction of fungi E.g. Griseofulvin, Ketoconazole
- ▶ **Antibacterial:** an agent that destroys or suppresses growth or reproduction of bacteria e.g. penicillin, Ampicillin, Amoxicillin
- ✓ **Bactericidal:** those which act primarily by **killing** bacteria
E.g. Penicillins, Cephalosporins, Aminoglycosides, Isoniazid and Rifampicin
- ✓ **Bacteriostatic:** act primarily by **arresting** bacterial multiplication E.g. Tetracyclines, Chloramphenicol

Phar-Terminologies Cont'd

- **Antiseptic:** substance that destroy microorganisms **on living tissue**
E.g. Iodine tincture
- **Disinfectant:** antimicrobial agents that are applied to **non-living** (inanimate) objects to destroy microorganisms
 - the process of which is known as **disinfection**
 - e.g. Formaldehyde, Phenol

Phar-Terminologies Cont'd

- ▶ **Antiflatulent (Carminative):** agent that relieves flatulence

E.g. Peppermint spirit, Simethicone

- ▶ **Antipyretic:** an agent that relieves or reduces fever

E.g. Paracetamol (Acetaminophen)

- ▶ **Antipruritic:** an agent that counteracts itching

HCl, other antihistamines

E.g. Hydroxyzine

- ▶ **Antispasmodic:** an agent that prevents or relieves spasms

Butylbromide

E.g. Hyoscine

Phar-Terminologies Cont'd

- **Antitussive:** an agent that suppresses coughing

E.g. Dextromethorphan HCl

- **Astringent:** an agent that causes precipitation of proteins and is effective in stopping the flow of blood or other secretion

- **Bronchodilator:** an agent which causes dilatation of the bronchi

E.g. Sulbutamol

Phar-Terminologies Cont'd

- **Contraindication:** any condition which renders a particular line of treatment improper or undesirable. E.g. **Thalidomide-pregnancy**
 - Sedative drug used to treat morning sickness



Phar-Terminologies Cont'd

- **Side effect:** **EXPECTED**, unwanted consequence other than that for which an agent is used E.g. Chlorpheniramine maleate –sedative
- **Adverse Drug Reaction (ADR):** **UNEXPECTED**, harmful or seriously unpleasant effects occurring at doses intended for therapeutic effect
 - call for reduction of dose or withdrawal of the drug and/or forecast hazard from future administration

Phar-Terminologies Cont'd

- **Drug interaction:** the action of one drug upon the effectiveness or toxicity of another (Increases or decreases)
 - Drug-drug, drug-food(diet) interaction
- **Degradation, deterioration:** Decomposition of a compound or becoming degraded by stages, exhibiting well-defined intermediate products(e.g. Oxidation, hydrolysis)

Phar-Terminologies Cont'd

- **Hygroscopic:** is the ability of a substance to attract water molecules from the surrounding environment through either **absorption** or **adsorption**
e.g. sugar, sulfuric acid, iodine

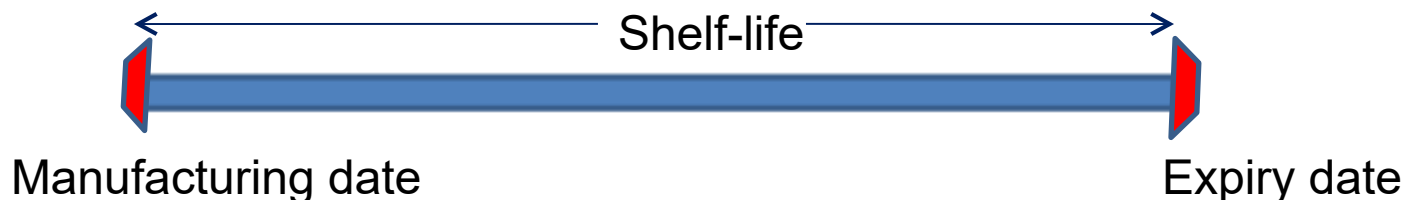
Phar-Terminologies Cont'd

- **Humectants:** a hygroscopic substance which when added to different dosage forms reduce the loss of moisture. E.g. glycerin, sorbitol
- **Emollient:** an agent that softens or soothes the skin E.g. Fixed oil, wax in cold cream
- **Expectorant:** an agent that promotes expectoration (coughing up and spitting out of material from the lungs, bronchi, and trachea)

E.g. Guaifenesin

Phar-Terminologies Cont'd

- **Expiry Date:** is the date before which the quality of a pharmaceutical remains acceptable for its intended use
- **Shelf life:** is the amount of time that a properly packed and stored product will last without undergoing chemical or physical changes Or
 - Length of time a product can be stored without deterioration



Phar-Terminologies Cont'd

- ▶ **Keratolytic:** an agent that promote keratolysis, e.g. salicylic acid
- ▶ **Placebo:** an inactive substance or preparation given to satisfy the patient's symbolic need for drug therapy
 - ▶ Used in controlled studies to determine the efficiency of medicinal substances. E.g. lactose tablet
- ▶ **Protectant:** an agent affording defense against harmful influence E.g. Sunscreen preparations

Introduction to Some Medical Terminologies

Medical Terminologies

- ▶ Words and phrases which have been developed to describe
 - ▶ the procedures
 - ▶ medications
 - ▶ instruments
 - ▶ anatomical structures, etc. used in the field of medicine
- ▶ It is the language of medicine
- ▶ Most medical terms derive from Latin or Greek
- ▶ Adheres to a set of fairly simple rules

How Are Medical Terms Constructed?

- Most medical terms consist of one or more parts
 - ▣ Root Words/Word Roots
 - ▣ Prefixes
 - ▣ Suffixes

Word roots:

- Contain the basic meaning of the term
- They usually, but not always, indicate the involved body part
 - part of the body condition that is being treated, studied or named by the term

Aden/o = gland

Cardi/o = heart

Dermat/o = skin

Enter/o = small intestines

gastr/o = stomach

Hemat/o = blood

Nephr/o = kidney

Neur/o = nerve

Pulmon/o = lungs

Suffixes

- Usually, but not always, indicate the procedure, condition, disorder, or disease
- A suffix always comes at the end of a word

E.g. Tonsill/o means tonsils. A suffix completes the word and tells what is happening to the tonsils

- Tonsillitis: tonsill (tonsil) + itis (inflammation)
- Tonsillectomy (tonsil) + ectomy(surgical removal)

➡ Suffixes Meaning Abnormal Condition

-**osis** means an abnormal condition or disease

- Gastrosis means any disease of the stomach

➡ Suffixes Related to Pathology/Disease condition

✓ -**algia** means pain

- Gastralgia, Arthralgia

✓ -**dynia** means pain

- Gastrodynia

✓ -**itis** means inflammation

- Gastritis, tonsillitis, bronchitis, neuritis

✓ -malacia means abnormal softening

- Arteriomalacia

- ✓ -**megaly** means enlargement
 - Hepatomegaly, cardiomegaly
- ✓ -**necrosis** means tissue death
 - Arterionecrosis
- ✓ -**sclerosis** means abnormal hardening
 - Arteriosclerosis
- ✓ -**stenosis** means abnormal narrowing
 - Aortic stenosis

➡ Suffixes Related to Procedures

- ✓ -**centesis** = surgical puncture to remove fluid
 - ✓ Abdominocentesis, Thoracentesis
- ✓ **ectomy** = surgical removal , resection, excision
 - Appendectomy, Tonsillectomy
- ✓ -graphy/gram = the process of recording a picture or record
 - Mammography
 - Arteriogram, Myelogram

Prefixes

- Usually, but not always, indicate the **location, time, number, status**
- A prefix always comes at the beginning of a word to change or add to its meaning
- Pre = before
 - Prenatal = time & events before birth
 - Precancerous = -----

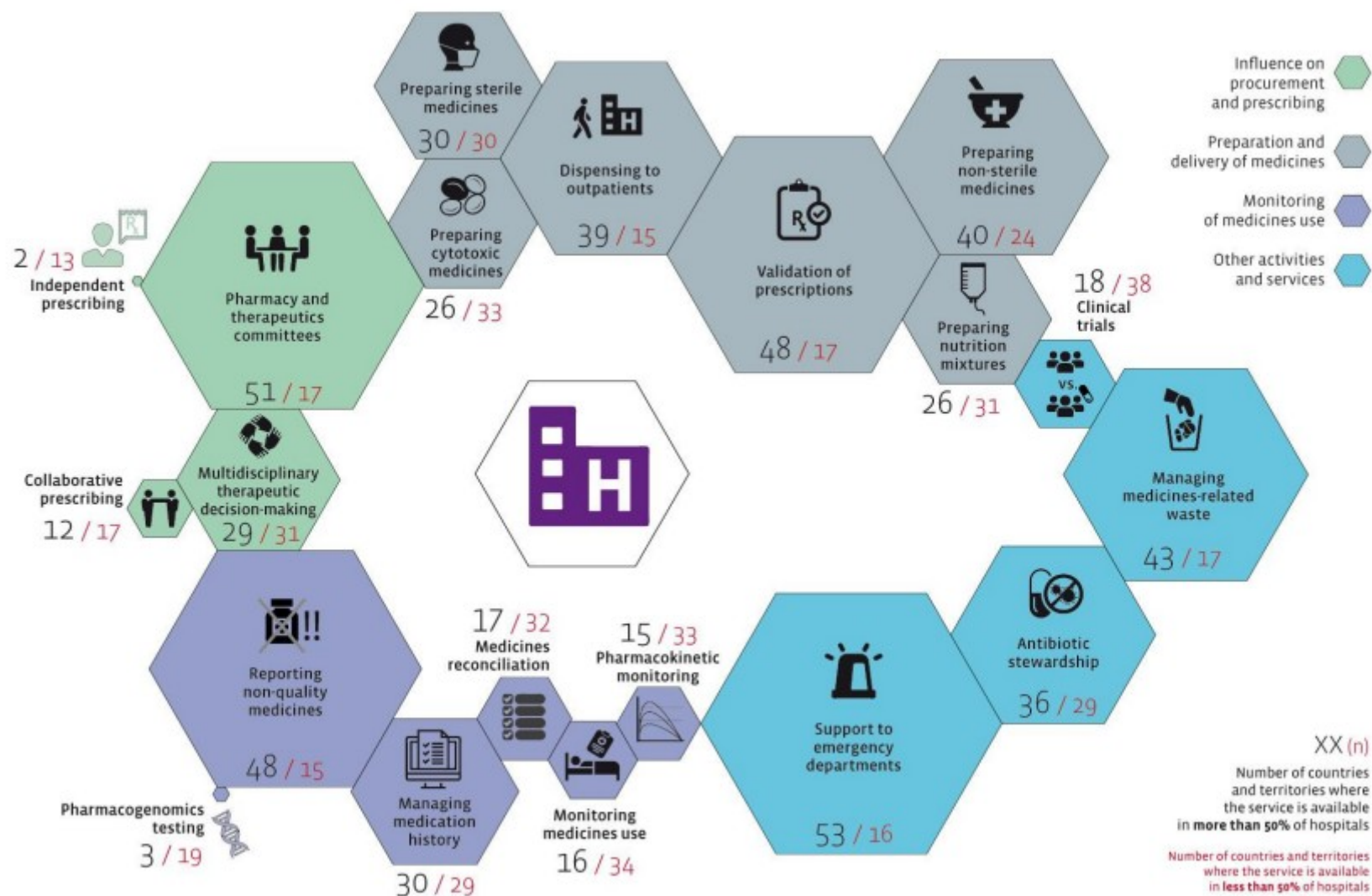
- Post- = After, Behind
 - Postnatal = time & event after birth
- Peri- = Surrounding (before , during and after)
 - Perinatal = time and events just before, during, & after birth
- Hyper- = Excessive, too much, above
 - *Hyperthyroidism*
- Hypo- = Deficient, too little, below
 - *Hypoglycemia*

- Number prefixes
 - Bi or Di = 2
 - Hemi or Semi = partial or half
 - Mono or Uni = 1
 - Multi = many
 - Null = none
 - Poly = many
 - Quad = 4
 - Tri = 3

Services provided by community pharmacy



Services provided by hospital pharmacy



Hospital Pharmacy

**TRUST ME, I'M A
PHARMACIST**

R_x

Outline

- Hospital Pharmacy
 - Meaning
 - Pharmaceutical services
- Aspect/Roles
 - Traditional Vs. New/Current
 - Rationale and Benefits
 - Clinical roles
 - Administrator/Manager roles
 - MDT Team Member roles
- Future
- Summary

Hospital Pharmacy: Meaning

- A **department** or a **service** in a hospital,
- Working under the direction of a professionally competent, legally **qualified pharmacist**,
- **Source**, from which all medications are supplied to the nursing units and other services.

Hospital Pharmacy



Meaning-Hospital Pharmacy...

- Where:
 - special **prescriptions** are **filled** for:
 - patients in the hospital,
 - ambulatory patients.
 - pharmaceuticals are **manufactured** in bulk,
 - narcotic and other prescribed drugs are **dispensed**,
 - injectable preparations should be **prepared** and **sterilized**, and
 - Medical supplies are often **stocked** and **dispensed**.

Hospital Pharmacy

- An **integral** part of a Hospital.
- **Linked/networked** to almost all facilities/ departments of the hospital,
 - **Operationally :**
 - Clinical departments and wards
 - Diagnostics departments- Laboratory services
 - **Structurally:-**
 - Senior management
 - Clinical services
 - Academic (Teaching hospitals)

Pharmaceutical services

- A hospital pharmacist is an **integral player** in the provision of pharmaceutical services
- Pharmaceutical services have the following prominent **components**:
 - The **procurement, distribution, and control** of all pharmaceuticals used within the facility.
 - The evaluation and dissemination of **comprehensive information** about drugs and their use to the institution's staff and patients.
 - The monitoring, evaluation, and assurance of the **quality** of drug use.

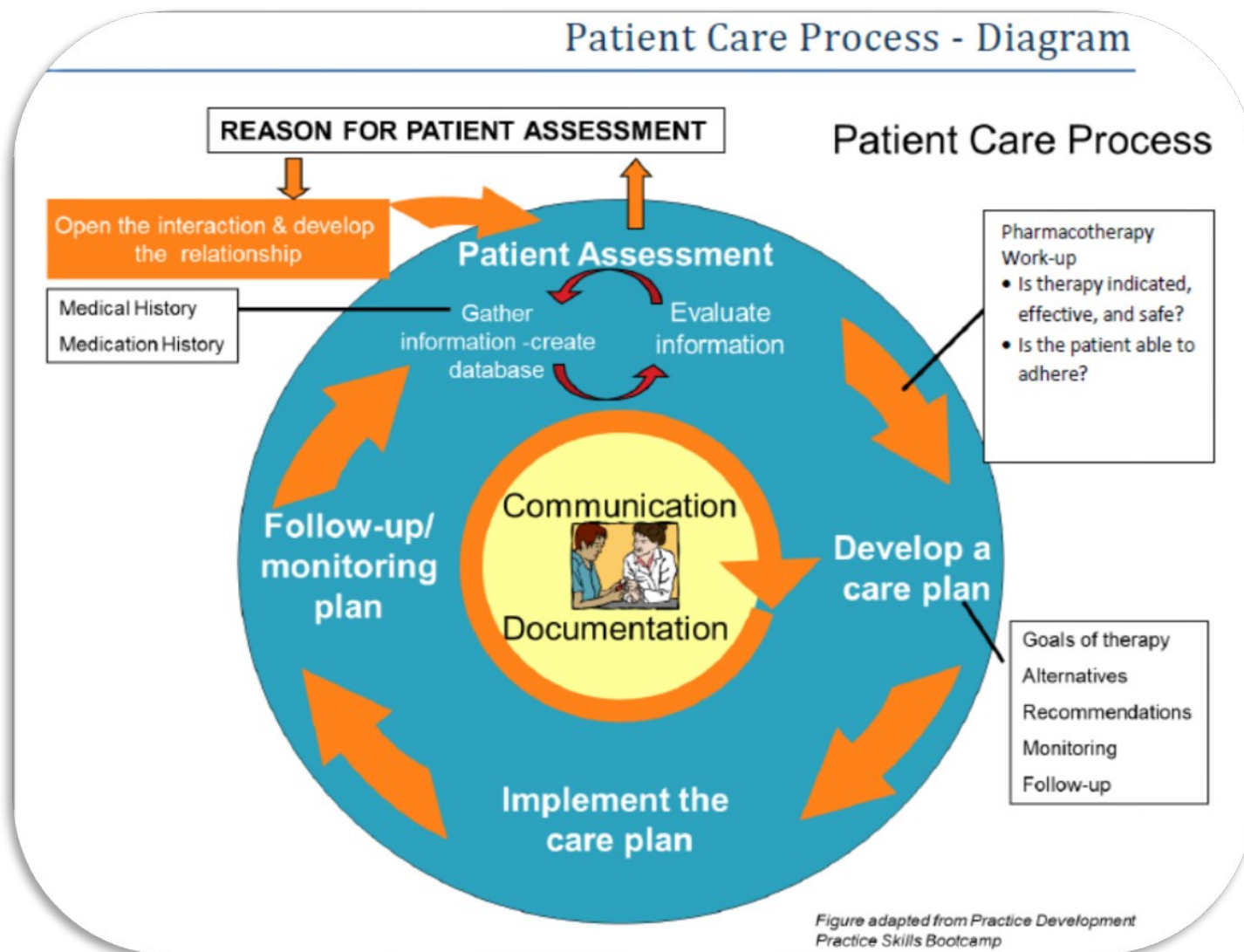
Pharmaceutical services...

- A pharmacist is **involved** in the execution of tasks and decision-making roles of:
 - **Drug selection**- an integral member of the team of experts
 - **Procurement**- a direct stakeholder & an integral member of the team of experts.
 - **Storage & Distribution**- a direct stakeholder
 - **Rational use**- a direct stakeholder & an integral member of the MDT.

Traditional Vs. Current role

- Traditionally, Pharmacists' roles were thought to be **restricted to** drug supply logistics:
 - manufacturing,
 - Distribution,
 - Storage,
 - Dispensing
- **Current/ new role** has been **expanded** to Clinical Pharmacy, in light of the patient-centered approach.
 - **Pharmaceutical Care!**

Patient Care Process - Diagram



Rationale for the 'Paradigm Shift'

“Drug related morbidity and mortality”

- 1987- 12,000 deaths and 15,000 hospital admissions reported to the FDA.
- Increased cost associated with drug-related problems (\$76.6 billion/annum in ambulatory care setting)*



*(*Johnson and Bootman, 1997; Ernst and Grizzle, 2001)*

Roles as a Clinical Pharmacist

- Prescription **monitoring**
- Medication history-taking and medicines **reconciliation**
- Prescribing **advice** to medical and nursing staff
- Medication errors and adverse drug reaction **reporting**

Roles as a Clinical Pharmacist...

- Patient education and counselling, including achieving **concordance**,
- Integrated medicines **management**,
- **Pharmacokinetics** and therapeutic drug level **monitoring**,
- **Education** and **training**,

Benefits of the changed role

- ↓ Medication **errors**
- ↓ Adverse drug **reactions**
- ↓ **Inappropriate** prescribing
- ↓ Medication **wastage**
- ↓ Hospital **stay**
- ↑ Drug interaction **detection**
- ↑ Cost **effective** prescribing
- ↑ **Quality** of life

Traditional vs. new role

Traditional

- Right **drug** to the right patient
- Knowledge of **drug** products and drug delivery system
- Patient instructed by **pharmacist**
- Pharmacists have their **own** world

New

- Focus moves from product to **patient**
- Knowledge of **disease** and **therapeutics**
- **Patient** ultimate decision maker
- **Cooperative** activity to improve quality of care

Traditional vs. new role...

Traditional

- Keeps record to satisfy **legal** requirements
- Serve patients at the **request** of prescribers
- **Inventory** generates revenue

New

- Keeps record to **evaluate** patient **outcomes**
- Serve patients **directly** and in cooperation
- Reimbursed for their **cognitive** services



Roles of a Hospital Pharmacist-Administrator/Manager

- A Pharmacist can have a managing and/or a leading role in **steering** a hospital pharmacy and a team of professionals.
- He/ She will be in **charge** of:
 - **Overseeing** the overall activities of the units,
 - **Supervising** the harmonized efforts of team members,
 - **Planning** the implementation of foreseen tasks and objectives,

Roles of a Hospital Pharmacist-Administrator/Manager...

- He/ She will be in charge of:
 - **Facilitating** the realization of outcomes/targets,
 - **Managing** the Pharmacy **team**,
 - Making sure the Pharmacy **Profession** is **upheld** amidst all operations and interactions with other health care professionals.

Roles of a Hospital Pharmacist- MDT leader/member

- A hospital Pharmacist's **role** in a Multi-disciplinary team is **integral**.
- **Various** clinical and quality-assurance **objectives** are met by teams and committees,
 - Drug and **Therapeutics** Committee
 - **Infection** Prevention
 - **Quality** Improvement and Assurance team
 - Medical **Equipment** Management

DTC- Pharmacist's responsibilities...

- In collaboration with the chairperson, **prepares agendas**, schedules **meetings** and notifies committee members.
- He/ She should **distribute** the **agenda** by ensuring all necessary documents requiring discussions or comments are attached,
- He/ She should **prepare minutes** of each meeting- summary of discussions and all action points clearly stated with the name of the responsible individual.
- He/ She should prepare and **circulate** all pertinent **materials** for meetings,

DTC- Pharmacist's responsibilities...

- He/ She should **conduct literature searches** and disseminate pertinent materials for meetings,
- He/ She should **acknowledge** the receipt of **ADR report** and other application formats.
- He/ She should **ensure** that the **decisions** taken by the committee are submitted to the Provost (CEO) for his/her notification.
- He/ She should **follow up** the action plans of the committee.

The 'Seven Star Pharmacist'

- Caregiver
- Communicator
- Decision-maker
- Teacher
- Life-long learner
- Leader
- Manager

WHO Consultative workshop on the Role of the Pharmacist, 1997

Recent Developments

- Conception and **implementation** of **APTS** (Auditable Pharmaceutical Transactions and Services): a transparent and documented provision of Pharmacy services.

- Pharmacy **automation**
- **Reportable** stock-status between PFSA and hospital



What more can be done?

Today's Pharmacists **can** engage in:

- Health **Promotion**
- Drug **information-**
 - Traditional Medicinal practice
- **Compounding** what can't be found!
- Chemo & Non-chemotherapy **admixture preparations**
- Small-scale **manufacturing**
- **Tailoring** selected **therapies** in Chronic illnesses

Summary

- Pharmacists are **integral professionals** in the Health-care settings (Hospitals).
- Their **roles** in the provision of medicines, supplies and medical equipment are **crucial** in the health-care delivery system.
- Both in education and practice, their role is **shifting** from the traditional 'behind the counter' to 'beyond' and is **patient-centered** rather than the product-oriented previous role.
- Pharmacists in the hospital setting **engage** in **planning, monitoring, evaluating, leading** and **managing** resources and operations.

New Services

- Medicine Use Reviews
- Minor Ailments Service – Including Chronic Disease Management
- Switching- POM to P
- Health Promotion / Screening
- Vaccination
- Generic Substitution

- **Manage Prescribed Medicines – Medicine Use Reviews**
ensure medicines are safe for the patient and taken correctly and minimise wastage.
- **Manage Chronic Conditions**
Better quality of life to patients improve outcomes and cost savings in secondary care.
- **Manage Common Ailments – Extend Equity to GMS Patients**
giving patients reassurance and advice, with or without the use of non-prescribed (OTC) medicines
- **Promote and Support Healthy Lifestyles**
helping people protect their own health - cholesterol testing, blood pressure testing, etc.
- **Generic Substitution.**
- **POM to P – Accessibility and Affordability.**

Future

What do you think?

Thank you